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An explicit lower bound for a homogeneous rational linear form in the logarithms of algebraic numbers. II

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Abstract. In this paper we establish an estimate for the modulus of a linear form of the type described in the title. This estimate is exponential with respect to the number of summands.

§ 1. Introduction

In this paper we substantially strengthen the results of the first part (see [1]). The exposition is practically independent of the first part. Moreover, we provide amplified comments on some of the topics discussed there.

As in [1], we consider only the homogeneous rational case of linear forms in logarithms of algebraic numbers, that is, expressions of the form

$$\Lambda = b_1 \ln \alpha_1 + \cdots + b_n \ln \alpha_n, \quad (1.1)$$

where $b_1, \dots, b_n \in \mathbb{Z}$, and $\alpha_1, \dots, \alpha_n \in \mathbb{K}^*$ are non-zero members of an algebraic number field $\mathbb{K} \subseteq \mathbb{C}$ with some fixed non-zero values of $\ln \alpha_j$.

The theory of linear forms in logarithms has a rather long history. There have recently been a number of substantial successes. The dependence of the estimate on A_j (the logarithmic heights of α_j), $|b_j|$ and the degree of $\mathbb{K}$ is virtually established. (The notation is described in more detail in the sections that follow.) All the estimates in recent publications are quite explicit and have numerous applications to other problems of number theory.

Thus the main interest shifted to a more precise description of how n enters into the estimate of Λ . Until recently, the main term here was of the form n^{2n} (this was first found in [2]). It was also noted that if $\alpha_1, \dots, \alpha_n$ are strongly independent, then there are estimates with main term n^n , and an extra factor $n!$ appears when one passes to the general case. In this paper, we understand by *strong independence* the validity of the equation

$$[\mathbb{K}(\sqrt{\alpha_1}, \dots, \sqrt{\alpha_n}) : \mathbb{K}] = 2^n. \quad (1.2)$$

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In the case when the α_j are strongly independent, there is a useful sharpening involving a parameter usually denoted by B . Here we can assume (see [3]) that

$$B = \max\{1, \max\{|b_j|A_j/A_n : 1 \leq j \leq n\}\}, \quad (1.3)$$

while in the general case we have

$$B^* = \max\{|b_1|, \dots, |b_n|\}. \quad (1.4)$$

As A_n increases, the formula (1.3) substantially strengthens the estimate of Λ . Various modifications of (1.3) appear in the literature.

In [1], an estimate without the factor n^n was obtained in the strongly independent case, and the general case was not considered.

The main results of the present paper are as follows.

1. The estimate of the linear form contains no factor n^n in all cases, that is, we eliminate (1.2) without losing accuracy. This is the novelty of this paper. It is interesting that the previous estimate of the linear form was worse in the general case because of the reduction to the strongly independent case. But in this paper it turns out that the situation is just the opposite: the case when (1.2) holds is in some sense the worst one.

2. The estimate of Λ contains no factor of the form $\ln(A_1 \dots A_{n-1})$, which was present in [1]. This is achieved using tools similar to those in [4].

3. As in [1], the main theorem imposes no individual lower bounds on the numbers A_j . The hypothesis of the theorem in [1] imposed a lower bound on their product Ω . We are now able to remove this restriction by using the results of [5], where an appropriate lower bound for Ω is proved.

4. As in [1], [3], the number B is understood to be given by (1.3), and this parameter is preserved in the general case.

We have listed the improvements on the previous paper. But the present paper also contains some losses, caused by the author's desire to make the main result more transparent and the exposition less complicated. These (compared to [1]) are as follows.

5. A number of parameters have disappeared from the exposition. These parameters enabled one to give more precise estimates in particular cases. We only retain a minimal set of the main parameters.

6. The scheme of arithmetic-analytic continuation is simplified at the expense of worsening the accuracy of the constants obtained. Therefore, if (1.2) holds, then the dependence on n in [1] is better than in the present paper. However, the constants obtained are quite explicit and are better than those previously known (see [4], for example).

After removing the factor n^n , we see that the exponential C^n becomes the main term in the dependence of the linear form Λ on n . It is hardly possible to remove this term. Hence the dependence of the estimates obtained on the parameters taken into account seems to be the best possible. On the other hand, the proof shows that there are new possibilities of strengthening the results.

§ 2. Statement of results

Throughout the paper we keep the following notation: $\mathbb{K}$ is an algebraic number field of degree D over $\mathbb{Q}$ embedded in $\mathbb{C}$. If $\mathbb{K} \subseteq \mathbb{R}$, we put $\varkappa = 1$, and otherwise $\varkappa = 2$. We are given numbers $\alpha_1, \dots, \alpha_n \in \mathbb{K}^*$ with absolute logarithmic heights $h(\alpha_j)$, $1 \leq j \leq n$ (see the definition in § 5). Let $\ln \alpha_1, \dots, \ln \alpha_n$ be arbitrary fixed non-zero values of the logarithms. Suppose that

$$A_j \geq \max\{D h(\alpha_j), |\ln \alpha_j|\}, \quad 1 \leq j \leq n. \quad (2.1)$$

In particular, we allow $\alpha_j = 1$, where $h(\alpha_j) = 0$.

We consider the linear form (1.1) with $b_1, \dots, b_n \in \mathbb{Z}$ and with the parameter B of (1.3). For brevity we put

$$\begin{aligned} \Omega &= A_1 \dots A_n, \\ C(n) &= C(n, \varkappa) = \frac{16}{n! \varkappa} e^n (2n+1+2\varkappa)(n+2)(4(n+1))^{n+1} \left(\frac{1}{2}en\right)^\varkappa, \\ C_0 &= \ln(e^{4.4n+7} n^{5.5} D^2 \ln(eD)), \quad W_0 = \ln(1.5eBD \ln(eD)). \end{aligned} \quad (2.2)$$

Theorem 2.1 (the main theorem). *If $\ln \alpha_1, \dots, \ln \alpha_n$ are linearly independent over $\mathbb{Z}$ and $b_n \neq 0$, then*

$$\ln |\Lambda| > -C(n)C_0W_0D^2\Omega. \quad (2.3)$$

Theorem 2.2. *Suppose that $\ln \alpha_1, \dots, \ln \alpha_r$ are linearly independent over $\mathbb{Z}$ for some r ($1 \leq r \leq n$). Assume (2.1) for $j \leq r$, and for $j > r$ put*

$$A_j \geq \max\{D h(\alpha_j), |\ln \alpha_j|, 0.16\}. \quad (2.4)$$

When $b_n = 0$, assume (2.4) for $j = n$ as well.

If $\Lambda \neq 0$, then

$$\ln |\Lambda| > -C(n) \max\{1, n/6\} C_0 W_0 D^2 \Omega. \quad (2.5)$$

Corollary 2.3. *Suppose that $\Lambda \neq 0$, A_j are given by (2.4), and B by (1.3). Then*

$$\begin{aligned} \ln |\Lambda| &> -C_1(n) D^2 \Omega \ln(eD) \ln(eB), \\ C_1(n) &= C_1(n, \varkappa) = \min \left\{ \frac{1}{\varkappa} \left(\frac{1}{2}en\right)^\varkappa 30^{n+3} n^{3.5}, 2^{6n+20} \right\}, \end{aligned} \quad (2.6)$$

where B may be replaced by B^ (see (1.4)).*

Remark 2.1. We include the factor D in the definition of A_j . Usually one regards D as a separate factor, which makes the expression similar to the absolute height and leads to the appearance of D in the estimate of Λ . But then the A_j become dependent on D as $\frac{1}{D} |\ln \alpha_j|$, and we see no major advantage in separating D . In all calculations we actually deal with A_j of the form (2.1).

Remark 2.2. The fixed lower bound for A_j in (2.4) suffices for all known algebraic numbers except roots of unity, which may lead to a small A . We also note that

lower bounds for A_j are necessary when we have no information on the linear independence of the logarithms. Otherwise a wrong improvement of the estimate may result, say, from splitting the component $b_j \ln \alpha_j$ with small A_j into several summands.

Theorem 2.2 does not exclude the possibility that some A_j are small. Moreover, if the field contains a root of unity α_0 of large degree ($\alpha_0 \neq \alpha_j, j \geq 1$), then we can formally add α_0 to the linear form with $b_0 = 0$. This improves the estimate of the linear form. This phenomenon (strengthening the estimate when the field contains roots of unity) was discovered by Feldman (see [6]).

Remark 2.3. We state Corollary 2.3 to simplify the application of our results in rough cases and to facilitate comparison with previous papers. Substantially better estimates are known for $n = 2$.

§ 3. Some notation and conventions

We use the vector notation introduced in [1]. There are different ways of enumerating the coordinates of a vector. We fix the notation for some of these:

- 1) $\bar{u} = (u_1, \dots, u_n)$,
- 2) $\tilde{u} = (u_1, \dots, u_{n-1})$,
- 3) $\bar{u}' = (u_0, \dots, u_n)$,
- 4) $\tilde{u}' = (u_0, \dots, u_{n-1})$.

For other (or irrelevant) orders of coordinates, we denote vectors by boldface letters, explaining the order of enumeration if necessary. (As a rule, we use subscripts $j = 1, \dots, J$.) A vector and its components are denoted by the same letter. For real vectors $\mathbf{u}, \mathbf{v}$ we write $\mathbf{u} \geq \mathbf{v}$ (or $\mathbf{u} > \mathbf{v}$) if the corresponding inequality holds for each coordinate.

Given a scalar function $f(x)$ of one scalar variable (possibly depending on other arguments) and given $\mathbf{u} = (u_1, \dots, u_J)$, we define a vector $f(\mathbf{u})$ by

$$f(\mathbf{u}) = (f(u_1), \dots, f(u_J)).$$

For example,

$$\ln \bar{\alpha} = (\ln \alpha_1, \dots, \ln \alpha_n), \quad \tilde{m}! = (m_1!, \dots, m_{n-1}!).$$

For a scalar function $f(x, y)$ and vectors $\mathbf{u}, \mathbf{v}$ we put

$$f(\mathbf{u}, \mathbf{v}) = \prod_{j=1}^J f(u_j, v_j).$$

For example,

$$\bar{\alpha}^{\bar{l}} = \prod_{j=1}^n \alpha_j^{l_j}.$$

We shall say that $\bar{l}$ is a *multipower* of $\bar{\alpha}$. To obtain the product of the coordinates of the vector $f(\mathbf{u})$ we write

$$f(\mathbf{u})^{\mathbf{1}} = \prod_{j=1}^J f(u_j)^1,$$

introducing the vector $\mathbf{1} = (1, \dots, 1)$. We assume that $0^0 = 1$.

Versions of the following obvious lemma are traditionally used in Baker's scheme.

Lemma 3.1. *Let $f(z, m)$, $g(z, m)$ be sets of polynomials (in z) with $\deg_z f = \deg_z g = m$, $m = 0, 1, \dots$. Then the polynomials $f(\mathbf{z}, \mathbf{m})$ are linearly expressible in terms of $g(\mathbf{z}, \mathbf{t})$, $\mathbf{t} \leq \mathbf{m}$, and vice versa.*

Following the modern notation (see [4]), we introduce the following differentiation operators for functions $f(\widetilde{Y'}) = f(Y_0, \dots, Y_n)$:

$$\partial_0 = \frac{\partial}{\partial Y_0}, \quad \partial_j = b_n Y_j \frac{\partial}{\partial Y_j} - b_j Y_n \frac{\partial}{\partial Y_n}, \quad 1 \leq j \leq n, \quad (3.1)$$

where the b_j are as in the linear form Λ (see (1.1)). Higher derivatives can be written using multipowers. For example,

$$\tilde{\partial}^{\tilde{m}} = \partial_1^{m_1} \dots \partial_{n-1}^{m_{n-1}}.$$

The following notation is related to these operators:

$$\tilde{\chi}(\bar{u}) = (\chi_1, \dots, \chi_{n-1}), \quad \chi_j = u_j b_n - u_n b_j, \quad 1 \leq j \leq n-1,$$

where $\bar{u}$ is necessarily a vector of the first type, and the resulting $\tilde{\chi}$ is a vector of the second type. The relation is given by

$$\tilde{\partial}^{\tilde{m}}(\bar{Y}^{\bar{u}}) = (\tilde{\chi}(\bar{u}))^{\tilde{m}} \bar{Y}^{\bar{u}}.$$

The set of positive (resp. non-negative) integers is denoted by $\mathbb{N}$ (resp. $\mathbb{N}_0$). The number of elements of a set $\mathcal{L}$ is denoted by $|\mathcal{L}|$.

Up to § 21, the paper is devoted to the proof of Theorem 2.1. Since $A_1, \dots, A_{n-1}$ occur symmetrically in the statement, we can assume without loss of generality that $A_1 \leq \dots \leq A_{n-1}$.

If $\varkappa = 1$, we have $|\alpha| = \pm \alpha \in \mathbb{K}$. Putting $\alpha'_j = |\alpha_j|$ and taking the real value of $\ln \alpha'_j$, $A'_j = A_j$, we get

$$|\Lambda| \geq |\Lambda'| = |b_1 \ln \alpha'_1 + \dots + b_n \ln \alpha'_n|.$$

Hence we can assume that $\alpha_j > 0$ and $\ln \alpha_j \in \mathbb{R}$. We must also consider the case $\alpha_j = \pm 1$ with non-zero $\ln \alpha_j$ because then $\ln \alpha'_j = 0$, which does not meet the assumptions of Theorem 2.1. But since $A_j \geq \pi$, we can simply omit the corresponding summand and thus improve the estimate. The case $\alpha_n = \pm 1$ is treated similarly to the case $b_n = 0$ in the proof of Theorem 2.2.

In what follows we encounter expressions depending on the parameters previously defined. For brevity we sometimes indicate only those parameters that are needed at the time. For example, we write C_0 , $C_0(n)$ or $C_0(n, D)$.

§ 4. Lattices

We present necessary facts from the basic geometry of numbers (see, for example, [7] and [8]). Let $\mathcal{V}$ be a vector space over $\mathbb{R}$ with a norm $\|\cdot\|$. A *lattice* is a discrete subgroup $M \subseteq \mathcal{V}$ (discrete means that for every R there are only finitely many vectors $\mathbf{v} \in M$ with $\|\mathbf{v}\| \leq R$). A lattice has a basis, in terms of which all

vectors of the lattice are integrally expressed. The number of vectors in a basis is called the *rank* of the lattice ($\text{rank } M$). Since M is discrete, $\text{rank}_{\mathbb{Z}} M = \text{rank}_{\mathbb{R}} M$.

Given a set $\mathcal{U}$ of vectors and a set $\mathcal{B} \subseteq \mathbb{R}$, we denote by $\langle \mathcal{U} \rangle_{\mathcal{B}}$ the set of (finite) linear combinations of vectors in $\mathcal{U}$ with coefficients in $\mathcal{B}$. For a finitely generated additive subgroup M to be a lattice, we must have $\text{rank } M = \dim \langle M \rangle_{\mathbb{R}}$. If $\langle M \rangle_{\mathbb{R}} = \mathcal{V}$, the lattice is called *complete*. Only complete lattices are studied in [7], but every lattice M is obviously complete in $\langle M \rangle_{\mathbb{R}}$.

Given a norm $\|\cdot\|$, we denote by $\mathcal{W}(\|\cdot\|)$ the unit ball, that is, the set of vectors with $\|\mathbf{v}\| \leq 1$. A number λ_k is the *kth successive minimum* of the set M with respect to the norm $\|\cdot\|$ if λ_k is the infimum of $\lambda > 0$ such that $\lambda \mathcal{W}(\|\cdot\|) \cap M$ contains k linearly independent vectors. Since a lattice is discrete, it has non-zero successive minima, which are achieved at some vectors. The successive minima are non-decreasing, and their number is equal to $\text{rank } M$.

If the space is Euclidean, the scalar product determines m -dimensional volumes by the formula

$$\text{Vol}_m \langle \mathbf{v}_1, \dots, \mathbf{v}_m \rangle_{[0;1)} = (\det(\mathbf{v}_i, \mathbf{v}_j)_{i,j=1,\dots,m})^{1/2},$$

where $\langle \mathbf{v}_1, \dots, \mathbf{v}_m \rangle_{[0;1)}$ is the parallelepiped spanned by the linearly independent vectors $\mathbf{v}_j$.

We note that one can work with norms on $\mathcal{V}$ which are not Euclidean. Moreover, there are different choices of the Euclidean metric which defines volumes. On the coordinate space, we usually take the standard metric (if not otherwise stated), but in what follows we also use other metrics. Let Q be the matrix of the Euclidean metric in some coordinate system, and let V be the matrix with rows formed by the coordinates of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$. Then

$$\text{Vol}_m \langle \mathbf{v}_1, \dots, \mathbf{v}_m \rangle_{[0;1)} = (\det(VQV^T))^{1/2}.$$

A solid $\Pi \subseteq \langle M \rangle_{\mathbb{R}}$ is called a *fundamental domain* of the lattice M if the sets $\Pi + \mathbf{v}$, $\mathbf{v} \in M$, form a decomposition of $\langle M \rangle_{\mathbb{R}}$. In particular, if $\mathbf{e}_1, \dots, \mathbf{e}_n$ is a basis of M , then $\Pi(M) = \langle \mathbf{e}_1, \dots, \mathbf{e}_n \rangle_{[0;1)}$ is called the *fundamental parallelepiped* of M . All fundamental domains have the same volume, which is called the *volume of the lattice* ($\text{Vol}_n M = \text{Vol}_n \Pi(M)$).

Proposition 4.1 (Minkowski's theorem on successive minima [7], Ch. 8, Theorem 5). *Suppose that we have a norm $\|\cdot\|$ in Euclidean space and a lattice M_n of rank n whose successive minima $\lambda_1, \dots, \lambda_n$ are achieved at vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$. Put $M'_n = \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{Z}}$, and let $[M_n : M'_n]$ be the index of M_n with respect to M'_n . Then*

$$\frac{1}{n!} 2^n [M_n : M'_n] \text{Vol}_n M_n \leq \lambda_1 \dots \lambda_n \text{Vol}_n \mathcal{W}(\|\cdot\|) \leq 2^n \text{Vol}_n M_n.$$

An unpleasant feature is that the minimal vectors may fail to form a basis: they generate a smaller lattice, which accounts for the factor $[M_n : M'_n]$.

Lemma 4.2 (Mahler [7], Ch. 5, Lemma 8). *Under the hypotheses of Proposition 4.1, the lattice M_n has a basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ such that*

$$\|\mathbf{e}_1\| \dots \|\mathbf{e}_n\| \text{Vol}_n \mathcal{W}(\|\cdot\|) \leq 2^{1-n} n! \text{Vol}_n M_n.$$

Let us explain the role of the factor $[M_n : M'_n]$. We consider a lattice M and arbitrary vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in M$. We put

$$M'_n = \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{Z}}, \quad M_n = M \cap \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{R}}.$$

Then M_n consists of vectors $\mathbf{v} \in M$ that are linearly dependent on the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ over $\mathbb{Z}$:

$$m_0 \mathbf{v} = m_1 \mathbf{v}_1 + \dots + m_n \mathbf{v}_n, \quad m_0, \dots, m_n \in \mathbb{Z}, \quad m_0 \neq 0.$$

It is known from the theory of groups that $[M_n : M'_n] \mathbf{v} \in M'_n$. Hence we may put $m_0 = [M_n : M'_n]$ in the formula above. Moreover, suppose that $\text{rank } M_n = \text{rank } M'_n = \nu > 0$. Then

$$[M_n : M'_n] = \frac{\text{Vol}_{\nu} M'_n}{\text{Vol}_{\nu} M_n}.$$

If $\nu = n$, then the relation between the vectors is unique up to an integer factor, and all m_j may be estimated in this way. We give a generalization of this assertion.

Lemma 4.3 ([8], Theorem 2). *Suppose that we have a norm $\|\cdot\|$ on a Euclidean space $\mathcal{V}$ and vectors $\mathbf{v}_j$, $0 < \|\mathbf{v}_j\| \leq A_j$, $1 \leq j \leq n$, with $M = \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{Z}}$ being a lattice of rank $r < n$. Then there are $n - r$ independent integer-valued relations ($m_{i,j} \in \mathbb{Z}$)*

$$m_{i,1} \mathbf{v}_1 + \dots + m_{i,n} \mathbf{v}_n = 0, \quad 1 \leq i \leq n - r,$$

which satisfy the inequality

$$\prod_{i=1}^{n-r} \max_j \{|m_{i,j}| A_j\} \leq \binom{n}{r}^{\frac{1}{2}} \frac{\mathcal{B}(\|\cdot\|, r)}{\text{Vol}_r M} A_1 \dots A_n. \quad (4.1)$$

If $r = n - 1$, they satisfy the inequality (we put $m_{1,j} = m_j$)

$$|m_j| A_j \leq \frac{\mathcal{B}(\|\cdot\|, r)}{\text{Vol}_r M} A_1 \dots A_n, \quad 1 \leq j \leq n, \quad (4.2)$$

where

$$\mathcal{B}(\|\cdot\|, r) = \sup \{ \text{Vol}_r \langle \mathbf{e}_1, \dots, \mathbf{e}_r \rangle_{[0;1]} : \mathbf{e}_j \in \mathcal{V}, \|\mathbf{e}_j\| \leq 1, 1 \leq j \leq r \}.$$

We note that A_j can be cancelled in (4.2), that is, the estimate for $|m_j|$ is independent of $\|\mathbf{v}_j\|$.

The inequality (4.2) suggests endowing the space $\mathbb{R}^n$ of vectors $(m_1, \dots, m_n)$ with a weighted max-norm

$$|\mathbf{m}|_A = \max \{|m_j| A_j : 1 \leq j \leq n\}. \quad (4.3)$$

Then (4.2) yields an estimate of $|\mathbf{m}|_A$. Weighted norms will also be useful in other cases.

We shall write linear forms as vectors, the value of a linear form $\mathbf{a}$ on a vector $\mathbf{x}$ being

$$\mathbf{a} \cdot \mathbf{x} = a_1 x_1 + \cdots + a_n x_n.$$

A norm $\|\cdot\|$ on a vector space defines the *conjugate norm* $\|\cdot\|^*$ on the space of linear forms. We have an unimprovable inequality

$$|\mathbf{a} \cdot \mathbf{x}| \leq \|\mathbf{a}\|^* \cdot \|\mathbf{x}\|.$$

The following sum-norm is conjugate to the max-norm (and vice versa):

$$\|x\| = |x_1| + \cdots + |x_n|.$$

The Euclidean norm is self-conjugate. If these norms contain a set $\mathcal{A}$ of weight coefficients, then the set $\mathcal{A}'$ of weight coefficients in the conjugate norms consists of their reciprocals.

Unless otherwise stated, we shall understand by $|\cdot|$ the max-norm and by $\|\cdot\|$ the sum-norm. If the weight coefficients are not indicated, they are assumed to be equal to 1.

Let vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ be linearly independent over $\mathbb{Z}$. Then the vectors $\mathbf{v}_i = m_{i,1}\mathbf{e}_1 + \cdots + m_{i,n}\mathbf{e}_n$, $i = 1, \dots, \nu \leq n$ ($m_{i,j} \in \mathbb{Z}$) can be extended to a basis in $\langle \mathbf{e}_1, \dots, \mathbf{e}_n \rangle_{\mathbb{Z}}$ if and only if the set of vectors $\mathbf{m}_i = (m_{i,1}, \dots, m_{i,n}) \in \mathbb{Z}^n$, $i = 1, \dots, \nu$ is *primitive*, that is, all $\nu \times \nu$ -minors of the matrix $(m_{i,j})$ are coprime.

The following lemma is a sharpened version of the estimate (4.2).

Lemma 4.4. *Suppose that $A_1, \dots, A_n > 0$ are weight coefficients with $A_1 \leq \cdots \leq A_{n-1}$, vectors $\mathbf{x}_i \in \mathbb{Z}^n$ are linearly independent, $\|\mathbf{x}_i\|_A \leq X_i$ ($i = 1, \dots, \nu \leq n$), and a vector $\mathbf{b} \in \mathbb{Z}^n$, $b_n \neq 0$, is linearly dependent on them over $\mathbb{Z}$. Then there is a non-trivial relation of the form*

$$m_0 \mathbf{b} = m_1 \mathbf{x}_1 + \cdots + m_\nu \mathbf{x}_\nu, \quad m_0, \dots, m_\nu \in \mathbb{Z}$$

such that

$$|m_0| \leq \frac{X_1 \cdots X_\nu}{A_1 \cdots A_{\nu-1} A_n},$$

$$|m_j| X_j \leq \nu |\mathbf{b}|_A \cdot \frac{X_1 \cdots X_\nu}{A_1 \cdots A_{\nu-1} A_n}, \quad 1 \leq j \leq \nu.$$

If the max-norms satisfy $|\mathbf{x}_i|_A \leq X_i$, $1 \leq i \leq \nu$, then one must introduce the factor $\nu^{\nu/2}$ on the right-hand side of both inequalities and remove the factor ν in the second.

Proof. Since $\mathbf{x}_i$ are linearly independent, we can find the coefficients m_i as follows. In the coordinate matrix $(x_{i,j})$ we choose a non-zero $\nu \times \nu$ -minor (with $j \in \mathcal{J}$). The resulting system of equations

$$m_1 x_{1,j} + \cdots + m_\nu x_{\nu,j} = m_0 b_j, \quad j \in \mathcal{J},$$

can be solved by Cramér's rule. This yields

$$m_0 = \Delta, \quad m_i = \Delta_i, \quad i = 1, \dots, \nu,$$

where $\Delta, \Delta_j \in \mathbb{Z}$, $\Delta \neq 0$. To give an upper estimate for $|\Delta_i|$, we multiply the j th column by A_j and, in the k th row, factor out the number $\nu|\mathbf{b}|_A$ if $k = i$ and the number X_k if $k \neq i$. Then the sum of absolute values of the remaining elements in every row does not exceed 1, and such a determinant cannot exceed 1. We similarly estimate $|\Delta|$. For the max-norm, Hadamard's inequality must be applied.

To complete the proof, note that since $b_n \neq 0$, there is a vector $\mathbf{x}_j$ with non-zero n th coordinate. Hence we can include n in the set $\mathcal{J}$. This proves the lemma.

To conclude the section, we note that the set $\mathcal{W}(\|\cdot\|)$ is a cube for the max-norm, an octahedron for the sum-norm (with a well-known formulae for the volume) and the usual ball for the Euclidean norm,

$$\text{Vol}_r \mathcal{W}(\|\cdot\|_2) = \frac{\pi^{r/2}}{(r/2)!},$$

where $(x)! = \Gamma(x+1)$ is the gamma-function.

§ 5. Heights

The well-known facts in this section are taken from [9], [10], and lower bounds for logarithmic heights from [5], [11].

Let $\mathbb{K}^{(\sigma)}$ be the embeddings of the field $\mathbb{K}$ in $\mathbb{C}$, $1 \leq \sigma \leq D$. We respectively have D Archimedean valuations of $\mathbb{K}$:

$$|\alpha|_\sigma = |\alpha^{(\sigma)}|, \quad 1 \leq \sigma \leq D.$$

Complex-conjugate embeddings correspond to equal valuations, but it is convenient to take them both into account. We enumerate simple ideals of $\mathbb{K}$ by indices $\sigma > D$ and define non-Archimedean valuations of $\mathbb{K}$ by the formula

$$|\alpha|_\sigma = (\text{Norm } \mathfrak{p})^{-k}, \quad k = \text{ord}_{\mathfrak{p}}(\alpha), \quad \mathfrak{p} = \mathfrak{p}_\sigma, \quad \sigma > D, \quad \alpha \in \mathbb{K}^*.$$

Then we have the product formula

$$\prod_{\sigma=1}^{\infty} |\alpha|_\sigma = 1, \quad \alpha \in \mathbb{K}^*.$$

Note that $|\alpha|_\sigma \neq 1$ for only finitely many σ .

The enumeration of valuations is convenient when we sum over all σ . To study linear forms in logarithms, we work with indices $\sigma = 1$ and $\sigma = 2$, which correspond to a pair of complex-conjugate fields. We omit the index $\sigma = 1$, thus identifying $\mathbb{K}$ with $\mathbb{K}^{(1)}$.

When studying the multiplicative group $\mathbb{K}^*$, we always assume (without explicit mention) that $\alpha \neq 0$.

Consider the logarithmic map of the multiplicative group $\mathbb{K}^*$ to the infinite-dimensional space $\mathbb{R}^\infty$ of vectors $\mathbf{x} = (x_\sigma \in \mathbb{R} : 1 \leq \sigma < \infty)$ given by the formula

$$x_\sigma = x_\sigma(\alpha) = \ln |\alpha|_\sigma, \quad 1 \leq \sigma < \infty.$$

This is clearly a homomorphism, and the image of $\mathbb{K}^*$ lies in the subspace $\mathcal{E} = \mathcal{E}(\mathbb{K}) \subseteq \mathbb{R}^\infty$ defined by the following conditions: $\mathbf{x} \in \mathcal{E}$ if

- a) almost all x_σ are equal to zero,
- b) $x_\sigma = x_\tau$ for complex-conjugate σ and τ ,
- c) the additive analogue of the product formula holds: $\sum_{\sigma=1}^{\infty} x_\sigma = 0$.

We call $\mathcal{E}$ the *logarithmic space*.

Remark. One usually replaces complex-conjugate pairs $x_\sigma = x_\tau$ by one number $2x_\sigma$, but we find it convenient to take all σ into account. Furthermore, the space $\mathcal{E}$ is infinite-dimensional, but this is merely a formal construction. We always work in some finite-dimensional coordinate subspace. For the needs of this paper, it suffices to use all Archimedean coordinates and the p -adic coordinates with $|\alpha_j|_\sigma \neq 1$ for at least one α_j .

We note that the logarithmic images $\mathbf{x}(\alpha)$ satisfy

$$x_\sigma \in \ln(\text{Norm } \mathfrak{p}_\sigma)\mathbb{Z}, \quad \sigma > D.$$

We define the *absolute logarithmic height* of a number $\alpha \in \mathbb{K}$ by

$$h(\alpha) = \frac{1}{D} \sum_{\sigma=1}^{\infty} \max\{0, \ln |\alpha|_\sigma\} = \frac{1}{2D} \sum_{\sigma=1}^{\infty} |\ln |\alpha|_\sigma|.$$

(The last equation holds by the product formula.) This defines a norm on the logarithmic space. Up to a factor, this is the sum-norm in $\mathcal{E}$, that is, $h(\alpha) = (2D)^{-1} \|\mathbf{x}(\alpha)\|$. This height is called absolute because it is independent of the field in which the number α may lie. If $\alpha \in \mathbb{Z}$ and $\alpha \neq 0$, then $h(\alpha) = \ln |\alpha|$. By Kronecker's theorem, the kernel of the logarithmic map is the group of roots of unity in the field $\mathbb{K}$. This group is finite and cyclic. The image of $\mathbb{K}^*$ is a lattice in $\mathcal{E}$.

We now construct the so-called *extended logarithmic space*. This construction was thoroughly described in great generality in [5]. In the previous space $\mathbb{R}^\infty$, which contains $\mathcal{E}$, we replace the $\mathbb{R}$ -components that correspond to the coordinates $\sigma = 1, \varkappa$ by $\mathbb{C}$ -components. The corresponding coordinates are $z_\sigma = x_\sigma + iy_\sigma$. The coordinates x_σ satisfy the conditions (a)–(c) above, and for y_σ we must have

- d) $y_\sigma = -y_\tau$ if σ and τ are complex conjugate.

The space obtained is called the *extended logarithmic space* $\mathcal{E}_\varkappa = \mathcal{E}_\varkappa(\mathbb{K})$. We can assume that $\mathcal{E} \subseteq \mathcal{E}_\varkappa$, where $\mathcal{E}$ consists of vectors with all y_σ equal to 0. There is a natural map of $\mathcal{E}_\varkappa$ to $\mathcal{E}$: the projection that annihilates all y_σ .

Let us describe how the group $\mathbb{K}^*$ is related to the space $\mathcal{E}_\varkappa$. To do this, we define the coordinates x_σ as before and y_σ by $y_\sigma = \text{Arg } \alpha^{(\sigma)}$. In other words, $z_\sigma = \text{Ln } \alpha^{(\sigma)}$, $\sigma = 1, \varkappa$. In constructing the corresponding lattice in $\mathcal{E}_\varkappa$, we take

all possible values of the logarithm. The resulting set of points of $\mathcal{E}_\varkappa$ is denoted by $M_\varkappa(\mathbb{K})$. If $\mathbf{x} \in M_\varkappa(\mathbb{K})$, then there is $\alpha \in \mathbb{K}^*$ such that

$$\begin{cases} \exp(x_\sigma) = |\alpha|_\sigma, & \sigma > \varkappa, \\ \exp(z_\sigma) = \alpha^{(\sigma)}, & \sigma \leq \varkappa. \end{cases}$$

This enables one to reconstruct α from its image. Thus the extended logarithmic map is not single-valued, but there is a single-valued inverse map of $M_\varkappa(\mathbb{K})$ onto $\mathbb{K}^*$ defined by the formulae above.

As noted in § 3, if $\mathbb{K} \subseteq \mathbb{R}$, then it suffices to prove the theorems only for positive numbers α_j with real values of the logarithms. We denote the corresponding objects by $\mathbb{K}^+$, $\mathcal{E}_\varkappa(\mathbb{K}^+)$ and $M_\varkappa(\mathbb{K}^+)$. On the other hand, we shall prove lemmas for the general case, when the assertions may differ from the case of $\mathbb{K}^+$.

To study the set $M_\varkappa(\mathbb{K})$, several norms on the space $\mathcal{E}_\varkappa$ were introduced in [5]. One of them is as follows. For $\vartheta > 0$ we set

$$H_\vartheta^*(\mathbf{x}) = \max \left\{ \max_{\sigma \leq \varkappa} |z_\sigma|, \frac{\vartheta}{2\varkappa} \sum_{\sigma > \varkappa} |x_\sigma| \right\}.$$

Then $H_\vartheta^*(\alpha) = H_\vartheta^*(\mathbf{x})$, where $\mathbf{x} = \mathbf{x}(\alpha) \in \mathcal{E}_\varkappa$ is a logarithmic image of the number α . Let us simplify this expression by summing over all $\sigma \geq 1$. We get

$$H_\vartheta(\alpha) = H_\vartheta(\mathbf{x}) = \max \left\{ \frac{D\vartheta}{\varkappa} h(\alpha), \max_{\sigma \leq \varkappa} |\ln \alpha^{(\sigma)}| \right\}.$$

We call $H_\vartheta(\alpha)$ the *extended logarithmic height* of α (with weight coefficient ϑ). Note that in the case of $\mathbb{K}^+$ we simply have $H_\varkappa(\alpha) = D h(\alpha)$. In [5] we considered the general case when $\varkappa$ assumes the values $1 \leq \varkappa \leq D$ and ϑ ranges from zero to infinity. In the present paper we consider, for simplicity, only $\varkappa \in \{1, 2\}$. Hence $\max\{|\ln \alpha^{(\sigma)}| : \sigma \leq \varkappa\} = |\ln \alpha|$.

Some previous papers considered a parameter

$$\frac{1}{E} = \max_j \left\{ \frac{1}{H_\varkappa(\alpha_j)} \max_{\sigma \leq \varkappa} |\ln \alpha_j^{(\sigma)}| \right\},$$

whose order corresponds to that of ϑ . The larger is E , the better are the estimates of linear forms. But the dependence of the norm $H_\vartheta(\alpha)$ on ϑ turns out to be useful because this is the norm for which we are able to prove a non-trivial estimate (in a form applicable to estimates of linear forms in logarithms) for successive minima of the lattice $M_\varkappa(\mathbb{K})$.

Proposition 5.1 ([5], Theorem 6.2). *Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be linearly independent vectors in $M_\varkappa(\mathbb{K})$. Put*

$$M_n = M_\varkappa(\mathbb{K}) \cap \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{R}}, \quad M'_n = \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{Z}}, \quad N = [M_n : M'_n].$$

Then

$$H_\vartheta^*(\mathbf{v}_1) \dots H_\vartheta^*(\mathbf{v}_n) \xi_0^* \max \left\{ 6, n-1, \frac{D}{\varkappa} \ln \left(4.64 \frac{D}{\varkappa} \right) \right\} > N,$$

$$\xi_0^* = \left(2 + \frac{1}{2\vartheta} \right)^n (e^{n/(1+4\vartheta)} n)^\varkappa.$$

To be consistent with the notation (2.1), we shall write $\vartheta = \varkappa$ and $H_\vartheta(\alpha) = H_\varkappa(\alpha)$. Taking into account that $\varkappa \leq 2$, we simplify the formula for ξ_0^* and present the estimate of Proposition 5.1 in a form which suffices for our applications (see the case $n = 1$ below):

$$\begin{aligned} A_1 \dots A_n \xi_0(n) \frac{D}{N} \ln(e^{4+n} D) &> 1, \\ \xi_0(0) &= 1, \quad \xi_0(1) = 1.5, \quad \xi_0(\nu) = 3.1^\nu \nu^\varkappa / \varkappa, \quad \nu \geq 2 \end{aligned} \quad (5.1)$$

(where $\ln \alpha_1, \dots, \ln \alpha_n$ are linearly independent over $\mathbb{Z}$). Replacing N by 1 and n by ν , we can also use (5.1) to estimate $A_1 \dots A_\nu$ ($1 \leq \nu \leq n$):

$$A_1 \dots A_\nu \xi_0(\nu) D \ln(e^{4+\nu} D) > 1. \quad (5.2)$$

We also note that the number $\ln(e^{4+n} D)$ can be replaced in (5.1), (5.2) by $C_0/2$ or by $W_0/2$ from (2.2). (See (6.3) for W_0 .)

The case $n = 1$ was previously considered in [11], where the following more precise estimate was obtained ([11], Corollary 4).

Proposition 5.2. *If $\alpha \in \mathbb{K}^*$ and $\deg \alpha = d > 1$, then*

$$\max\{d h(\alpha), |\ln \alpha|\} \geq \frac{8}{9d \ln(2.2d)}.$$

Let us rewrite the inequality of Proposition 5.2 as

$$\frac{1}{A_j} \leq 1.5D \ln(eD), \quad (5.3)$$

which is also valid for $D = 1$ (since we have either $h(\alpha) \geq \ln 2$ or $\alpha = \pm 1$). This explains the value $\xi_0(1) = 1.5$.

In 1933, Lehmer obtained the number $\alpha = 1.176\dots$ of degree $d = 10$, which is a root of

$$x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1 = 0$$

and has the minimal known logarithmic height $d h(\alpha) = 0.162\dots$ among all algebraic numbers other than roots of unity. This explains the choice of the constant 0.16 in (2.4).

Lemma 4.2 gives the assertion for general lattices and norms, containing the parameters $\mathcal{B}(\|\cdot\|, r)$ and $\text{Vol}_r M$. Only their ratio is important, and we can arbitrarily choose the Euclidean norm that defines the volumes. We now state the estimates of $\mathcal{B}(\|\cdot\|, r)$, $\text{Vol}_r M$ for lattices in $\mathcal{E}_\varkappa(\mathbb{K})$ to be used in what follows. For the Euclidean norm we take

$$\|\mathbf{v}\|_2^2 = \sum^{(*)} y_\sigma^2 + \sum_{\sigma=1}^{\infty} x_\sigma^2, \quad (5.4)$$

where $\sum^{(*)}$ means that we take only one coordinate from each pair of complex-conjugate coordinates (this component is absent for $\mathbb{K}^+$).

Lemma 5.3. *Let $\mathcal{E}_{\mathcal{K}}(\mathbb{K})$ be endowed with the norm $\|\cdot\| = H_{\mathcal{K}}(\cdot)$, and let the scalar product be induced by the norm (5.4) with k terms y_{σ}^2 in $\Sigma^{(*)}$. Then we have the estimate*

$$\mathcal{B}(\|\cdot\|, r) \leq (k+2)^{r/2}.$$

For every r -dimensional lattice $M \subseteq M_{\mathcal{K}}(\mathbb{K})$ with $\mathbb{K} \neq \mathbb{Q}$ we have

$$\text{Vol}_r M \geq \frac{\pi^{r/2}}{(r/2)!} \left(\frac{9}{8} D^{3/2} \ln(eD) \right)^{-r}. \quad (5.5)$$

In the case $\mathbb{K} = \mathbb{Q}$ we have

$$\text{Vol}_r M \geq P_1 \dots P_r, \quad (5.6)$$

$P_1 = \ln 2$, $P_2 = \ln 3$, $\dots$, $P_9 = \ln 23$, $P_{10} = \pi$, $P_{11} = \ln 29$, $\dots$ (the logarithms of successive primes), and in the case of $\mathbb{Q}^+$ the number π must be removed from the sequence.

In the case when $\mathbb{K} \neq \mathbb{Q}$, $k \leq 1$, as well as in the case of $\mathbb{Q}^+$, $k = 0$, we have

$$\frac{\mathcal{B}(\|\cdot\|, r)}{\text{Vol}_r M} \leq 2.2 \left(\frac{r}{2} \right)! \left(\frac{3}{\pi} \right)^{r/2} \left(\frac{9}{8} D^{3/2} \ln(eD) \right)^r. \quad (5.7)$$

We note that (5.7) differs from (5.5) by the factor 2.2, whose only role is to make the right-hand side of (5.6) less than that of (5.5) for $D = 1$ and $r < 4$.

The proof is given in § 22.

We recall the notion of height in vector $\mathbb{K}$ -spaces. The *local* and *absolute logarithmic heights* of a set $\{\beta_j \in \mathbb{K} : j \in \mathcal{J}\}$ are defined by

$$h^{(\sigma)}(\{\beta_j\}) = \ln \max\{|\beta_j|_{\sigma} : j \in \mathcal{J}\}, \quad \sigma \geq 1,$$

$$h(\{\beta_j\}) = \frac{1}{D} \sum_{\sigma=1}^{\infty} h^{(\sigma)}(\{\beta_j\}).$$

The height $h(\cdot)$ is projective, that is, $h(\{\lambda\beta_j\}) = h(\{\beta_j\})$ for $\lambda \in \mathbb{K}^*$. (This follows from the product formula.)

In what follows we consider the values of polynomials of several variables at a point $\bar{\alpha}$. Here the set $\{\beta_j\}$ will be formed by the numbers $\bar{\alpha}^{\bar{l}}$, where $\bar{l}$ runs over a set $\mathcal{L} \subseteq \mathcal{W} \subseteq \mathbb{R}^n$ of multipowers. The following notation will be convenient for estimates. Given a vector $\bar{\alpha} \in (\mathbb{K}^*)^n$ and a set $\mathcal{W} \subseteq \mathbb{R}^n$, we put

$$h^{(\sigma)}(\bar{\alpha}, \mathcal{W}) = \sup\{\bar{w} \cdot \ln |\bar{\alpha}|_{\sigma} : \bar{w} \in \mathcal{W}\}, \quad \sigma \geq 1,$$

$$h(\bar{\alpha}, \mathcal{W}) = \frac{1}{D} \sum_{\sigma=1}^{\infty} h^{(\sigma)}(\bar{\alpha}, \mathcal{W}).$$

Lemma 5.4 ([1], Lemma 3.2). *The height $h(\bar{\alpha}, \mathcal{W})$ has the following properties:*

- (i) if $\mathcal{V} \subseteq \mathcal{W}$, then $h(\bar{\alpha}, \mathcal{V}) \leq h(\bar{\alpha}, \mathcal{W})$;
- (ii) $h(\bar{\alpha}, r\mathcal{W}) = r h(\bar{\alpha}, \mathcal{W})$, $r > 0$;
- (iii) $h(\bar{\alpha}, \mathcal{V} + \mathcal{W}) = h(\bar{\alpha}, \mathcal{V}) + h(\bar{\alpha}, \mathcal{W})$;
- (iv) $h(\bar{\alpha}, \{\bar{v}\}) = 0$, $\bar{v} \in \mathbb{R}^n$;
- (v) $h(\bar{\alpha}, \mathcal{W} + \bar{v}) = h(\bar{\alpha}, \mathcal{W})$, $\bar{v} \in \mathbb{R}^n$;
- (vi) $h(\bar{\alpha}, \mathcal{W})$ is independent of the field containing the numbers α_j .

§ 6. Liouville inequalities

In this section we prove Theorem 2.1 for $n = 1$ as well as for comparatively small B . In the theory of Diophantine approximations, inequalities that follow directly from the product formula are usually called *Liouville inequalities*. The parameter B in (1.3) turns out to be very convenient in estimates of this kind.

Lemma 6.1 ([1], Lemma 13.1). *Suppose that $\Lambda \notin 2\pi i\mathbb{Z}$ in the notation of Theorem 2.1. Then*

$$|e^\Lambda - 1| \geq 2^{1-D} e^{-nBA_n}.$$

Since there are no formal lower bounds for parameters taken into account in the estimate of Λ , we now must verify some assertions that were previously obvious.

Lemma 6.2. *Theorem 2.1 is true for $|\Lambda| \geq 0.001$.*

Proof. By (5.2), $\xi_0(n)D\Omega C_0 \geq 1$. Hence the right-hand side of (2.3) still contains a sufficiently large factor. This proves the lemma.

We may thus assume in what follows that $|\Lambda| < 0.001$. Then Lemma 6.1 yields the inequality

$$|\Lambda| \geq \frac{1}{2} |e^\Lambda - 1| \geq \exp(-D \ln 2 - nBA_n).$$

We now consider the case when the Liouville estimate is worse than (2.3). Then

$$-D \ln 2 - nBA_n \leq -C(n)D^2\Omega C_0 W_0,$$

or, on account of (5.2),

$$nBA_n \geq C(n)D^2\Omega C_0 W_0 - D \ln 2 \geq 0.99C(n)D^2\Omega C_0 W_0.$$

Cancelling nA_n , we get

$$B \geq 0.99 \frac{1}{n} C(n) D^2 A_1 \dots A_{n-1} C_0 W_0. \quad (6.1)$$

Here the exponential part $C(n)$ is of the form $(4e^2)^n$. In particular, using estimate (5.2) for the factor $DA_1 \dots A_{n-1} C_0/2$, we get $B \geq e^{2n+8} D$. This enables us to use (5.1), (5.2) with $\ln(e^{4+\nu} D)$ replaced by $W_0/2$. Hence (6.1) implies that

$$B \geq 1.98C(n)C_0 D/n\xi_0(n-1), \quad (6.2)$$

and W_0 satisfies the inequality

$$W_0 \geq 2.25n + 1.5 \ln n + 13 + 2 \ln D, \quad (6.3)$$

which is proved directly for n small and asymptotically for n large, since the base of the exponential of n in (6.2) satisfies $\ln(4e^2/3.1) > 2.25$.

Moreover, (6.1) yields the estimate

$$A_1 \dots A_{n-1} \leq B \left(0.99 \frac{1}{n} C(n) D^2 C_0 W_0 \right)^{-1}. \quad (6.4)$$

Thus we may always assume that inequality (6.1) holds (otherwise Theorem 2.1 is true by the Liouville estimate). Hence inequalities (6.2)–(6.4) also hold.

Lemma 6.3. *Theorem 2.1 is true for $n = 1$.*

Proof. If $n = 1$, we have

$$|\Lambda| = |b_n \ln \alpha_n| \geq |\ln \alpha_n|.$$

Hence we may assume that $B = 1$, and therefore Theorem 2.1 follows from the Liouville estimate.

To conclude the section, we note that the dependence on D is optimal in the calculations above and in the estimates that follow. This dependence was first established for arbitrary n in [2].

§ 7. The inductive assumption

Except for §§ 21, 22, this paper is devoted to the proof of Theorem 2.1. The proof is by induction on n (the number of logarithms in the linear form Λ). The base of the induction ($n = 1$) is proved in § 6, and we assume that $n \geq 2$. We may also assume that the set b_j is primitive. Otherwise, cancelling the common factor, we decrease B and improve the estimate of the linear form.

Let us explain one more assumption introduced in this section. Using properties of the logarithmic function, we can decrease the number of logarithms in the linear form Λ :

$$\Lambda = b_1 \ln \alpha_1 + \cdots + b_n \ln \alpha_n = (m_1 \ln \vartheta_1 + \cdots + m_r \ln \vartheta_r)/m_0. \quad (7.1)$$

For example, write $\Lambda = \ln \vartheta$, $\vartheta = \bar{\alpha}^{\bar{b}}$. This usually increases the heights of ϑ_i and worsens the estimate. However, if the increase in heights is small, we can prove the theorem directly from the inductive assumption for $r < n$. We now describe this situation on the basis of the arguments in [4].

Recall that in Theorem 2.1 we assume that $\ln \alpha_j$ are linearly independent over $\mathbb{Z}$. Suppose that for some r ($1 \leq r < n$) there is a set of vectors

$$\overline{Z}_i = (z_{i,1}, \dots, z_{i,n}) \in \mathbb{Z}^n, \quad 1 \leq i \leq r,$$

such that

- (i) $\overline{Z}_1, \dots, \overline{Z}_r$ are linearly independent over $\mathbb{Z}$;
- (ii) there is a primitive set $m_0, \dots, m_r \in \mathbb{Z}$ with $m_0 \neq 0$ such that

$$m_0 \bar{b} = m_1 \overline{Z}_1 + \cdots + m_r \overline{Z}_r; \quad (7.2)$$

(iii) for the sum-norm with weights A_j (as in the hypothesis of Theorem 2.1) we have

$$\|\overline{Z}_1\|_A \cdots \|\overline{Z}_r\|_A \leq \frac{1}{2} (4e^2)^{n-r} A_1 \cdots A_n. \quad (7.3)$$

Lemma 7.1. *If conditions (i)–(iii) hold, then the validity of Theorem 2.1 for n follows from its validity for $r < n$.*

Proof. Define algebraic numbers ϑ_i and their logarithms by

$$\ln \vartheta_i = \overline{Z}_i \cdot \ln \bar{\alpha} = z_{i,1} \ln \alpha_1 + \cdots + z_{i,n} \ln \alpha_n, \quad 1 \leq i \leq r.$$

Then (7.1) obviously holds with the coefficients m_i in (7.2), and the logarithms $\ln \vartheta_i$ are linearly independent over $\mathbb{Z}$. By the definition of A_j and properties of the height we have

$$H_{\mathcal{K}}(\vartheta_i) \leq A'_i = \|\overline{Z}_i\|_A, \quad 1 \leq i \leq r.$$

Since $b_n \neq 0$, there is a non-zero number $z_{i,n}$, $1 \leq i \leq r$, with the corresponding $m_i \neq 0$. We may assume without loss of generality that $z_{r,n} \neq 0$, $m_r \neq 0$. Then $A'_r \geq |z_{r,n}| A_n \geq A_n$.

In analogy with the notation of § 2, we put

$$B' = \max\{|m_i| A'_i / A'_r : 1 \leq i \leq r\}, \quad \Omega' = A'_1 \dots A'_r.$$

Let us estimate the number B' using Lemma 4.3 and inequality (7.3). We have

$$B' A'_r = \max\{|m_i| A'_i : 1 \leq i \leq r\} \leq \frac{r B \Omega'}{A_1 \dots A_{r-1}} \leq \frac{1}{2} r B (4e^2)^{n-r} \frac{A_1 \dots A_n}{A_1 \dots A_{r-1}}.$$

Cancelling $A'_r \geq A_n$ and applying (5.2) to $1/(A_1 \dots A_{r-1})$, we see on account of (6.4) that

$$B' \leq \frac{1}{2} r B (4e^2)^{n-r} \xi_0(r-1) D \ln(e^{n+4} D) \frac{2nB}{C(n) D^2 C_0 W_0} \leq B^2.$$

Using Lemma 4.3 and inequalities (7.3), (5.2), (6.4), we similarly estimate m_0 :

$$\begin{aligned} |m_0| &\leq \|\overline{Z}_1\|_A \dots \|\overline{Z}_r\|_A \frac{1}{A_1 \dots A_{r-1} A_n} \\ &\leq \frac{1}{2} (4e^2)^{n-r} A_1 \dots A_{n-1} \xi_0(r-1) D \ln(e^{n+4} D) \\ &\leq \frac{1}{2} (4e^2)^{n-r} \frac{2nB}{C(n) D^2 C_0 W_0} \xi_0(r-1) D \ln(e^{n+4} D) \leq B. \end{aligned}$$

We now apply Theorem 2.1 to the form $m_0 \Lambda$ from (7.1) with the numbers ϑ_i and m_i . We get (again on account of (7.3))

$$\begin{aligned} -\ln |\Lambda| &\leq \ln |m_0| + C(r) D^2 \Omega' C_0 \ln(1.5eB'D \ln(eD)) \\ &\leq \ln B + C(r) \frac{1}{2} (4e^2)^{n-r} D^2 \Omega C_0 \ln(1.5eB^2 D \ln(eD)) \\ &\leq W_0 + 4e^2 C(n-1) D^2 \Omega C_0 W_0 \leq C(n) D^2 \Omega C_0 W_0. \end{aligned}$$

This proves Theorem 2.1 for n in the case under study. We have used the easily verifiable inequality

$$\frac{C(n)}{C(n-1)} > 4e \left(1 + \frac{1}{n}\right)^{n+2} > 4e^2 \left(1 + \frac{1}{n}\right).$$

In what follows we thus assume that the hypotheses of Lemma 7.1 do not hold, that is, for any r ($1 \leq r \leq n$) there are no vectors $\overline{Z}_i$ with properties (i)–(iii). This constitutes a difference between our inductive scheme and that of Baker and Wüstholz, who maximally reduce the linear form Λ and work with the truncated form. The scheme we suggest economizes notation.

We note that this is one of the points of the proof where we can use weight coefficients to preserve the dependence of the estimate of Λ on B from (1.3), not on B^* from (1.4).

The inductive assumption will be used only at the end of the proof, in § 20.

We also note that the idea of reduction was suggested in [12] (1949) to pass from $n = 3$ to $r = 2$. Of course, the number of variables in the linear form Λ should be decreased when possible. The main difficulty concerns using the information that the form is not simplified. This was recently seen to be a key condition in the zero estimates (see § 19).

§ 8. Generalized binomial polynomials

This is an auxiliary section. Here we present information on basis polynomials with required arithmetic properties. We put

$$\Delta(z, 0) = 1, \quad \Delta(z, k) = \frac{1}{k!} z(z+1) \dots (z+k-1), \quad k = 1, 2, \dots$$

For a fixed $H \in \mathbb{N}$ we put

$$\Delta(z, l, H) = (\Delta(z, H))^\lambda \Delta(z, h), \quad l = \lambda H + h, \quad 0 \leq h \leq H, \quad l \in \mathbb{N}_0.$$

We call $\Delta(z, l, H)$ a *generalized binomial polynomial*.

We note that if $l > 0$ is divisible by H , then there are two representations $l = \lambda H + H = (\lambda + 1)H$, but $\Delta(z, l, H)$ is easily seen to be independent of the choice of representation. The first representation with smaller λ is more convenient in estimates. Clearly, $\deg \Delta(z, l, H) = l$. These polynomials are modifications of similar ones introduced by Feldman and Baker. By $\Delta^{(m)}(cz, l, H)$ we mean the derivative of $\Delta(cz, l, H)$ with respect to z .

The following property of the polynomials $\Delta(z, l, H)$ is proved in [13].

Lemma 8.1. *Suppose that $H, L_0 \in \mathbb{N}$; $m_0, L' \in \mathbb{N}_0$; $L_0 = L'H + H_1$, $0 \leq H_1 \leq H$; $d_0(m_0)$ is the lowest common denominator of the numbers*

$$\frac{1}{m!} \Delta^{(m)}(x, l, H), \quad x \in \mathbb{Z}, \quad 0 \leq l \leq L_0, \quad 0 \leq m \leq m_0.$$

Then

$$d_0(m_0) \leq H!^{L'} H_1! \left(2.8361 e^{\frac{m_0 + L' + 1}{L_0}} \right)^{L_0}, \quad (8.1)$$

and also

$$d_0(m_0) \leq \exp(1.03883 H m_0). \quad (8.2)$$

We note that the Feldman-type estimate (8.1) contains a power of m_0 and is somewhat more precise than the Baker-type estimate (8.2). For simplicity we use (8.2), which is more adapted to estimates of other components.

In what follows we consider $x \in \frac{1}{K}\mathbb{Z}$, $K = 2^t$. To avoid fractional values of the polynomials, we consider $\Delta(Kx, l, H)$. This is also useful because we need not multiply $\Delta^{(m)}(x, l, H)$ by

$$\Pi(K, l) = \prod_{p|K} p^{[l/(p-1)]}$$

(p is a prime). We note that $\Pi(K, l) = 2^l$ for $K = 2^t$.

Besides Lemma 8.1, [13] contains an estimate of $|\Delta^{(m)}(z, l, H)|$. Here we obtain an improved estimate of this number.

Lemma 8.2. *Suppose that $H \in \mathbb{N}$, $z, c \in \mathbb{C}$, $c \neq 0$. Then*

$$\left| \frac{1}{\tau!m!} c^{-m} \Delta^{(m+\tau)}(cz, l, H) \right| = \delta(l, m, \tau) \left(e \left(1 + |c| \frac{|z|+1}{H} \right) \right)^l, \quad (8.3)$$

$$\sum_{l=0}^{\infty} \sum_{m+\tau \leq l} \delta(l, m, \tau) \leq \frac{e}{e-1} e^{H/e},$$

where the numbers $\delta(\cdot)$ also depend on c, z, H .

Proof. The polynomial component in the left-hand side of the first line of (8.3) does not exceed $(|cz| + H - 1)^{l'} |c|^{\tau} 1^m$, where $l' = l - m - \tau$. The factorials produced by differentiation and the number of summands are bounded above by $\frac{(m+\tau)!}{m!\tau!} \binom{l}{m+\tau}$. Summing over $m + \tau \leq l$ (that is, $m + \tau + l' = l$), we get

$$\sum_{m+\tau+l'=l} \frac{l!}{m!\tau!l'!} (|cz| + H - 1)^{l'} |c|^{\tau} 1^m = ((|cz| + H - 1) + |c| + 1)^l.$$

Factoring out the number $(H/e)^l$, we get the first line of (8.3).

Summing over $l = H\lambda + h \geq 0$ (or over $\lambda \geq 0$, $0 \leq h < H$) and taking the definition of $\Delta(z, l, H)$ into account, we obtain the inequality

$$\sum_{l=0}^{\infty} \left(\frac{H}{e} \right)^l \frac{1}{H!^{\lambda}} \frac{1}{h!} = \sum_{\lambda=0}^{\infty} \left(\frac{1}{H!} \left(\frac{H}{e} \right)^H \right)^{\lambda} \sum_{h=0}^{H-1} \frac{1}{h!} \left(\frac{H}{e} \right)^h \leq \sum_{\lambda=0}^{\infty} \left(\frac{1}{e} \right)^{\lambda} e^{H/e},$$

which yields the second line of (8.3).

We prove a technical estimate which was not distinguished in [1] as a separate assertion.

Lemma 8.3. *Suppose that $\mathbf{x}, \mathbf{y} \in \mathbb{C}^J$, $\mathbf{m} \in \mathbb{N}_0^J$, $\|\mathbf{m}\| \leq M \leq \mu$, $\mu \geq 1$, and $\gamma_0 \geq \|\mathbf{m}\| + \|\mathbf{x}\| + \|\mathbf{y}\| > 0$. Then*

$$\mathcal{D} = \sum_{\mathbf{t}} \frac{1}{\mathbf{t}!^1} |\mathbf{x}^{\mathbf{t}}| \cdot |\Delta(\mathbf{y}, \mathbf{m})| \leq \frac{1}{M!} \gamma_0^M, \quad \left(\frac{\gamma_0}{\mu} \right)^{\mu-M} \mathcal{D} \leq \frac{1.1}{\sqrt{2\pi\mu}} \left(\frac{e\gamma_0}{\mu} \right)^{\mu}. \quad (8.4)$$

(Here the sum runs over $\mathbf{t} \in \mathbb{N}_0^J$, $\|\mathbf{t}\| = M - \|\mathbf{m}\|$, $\|\cdot\|$ denotes the sum-norm, and $\mathbf{1} = (1, \dots, 1) \in \mathbb{Z}^J$. For the meaning of $\mathbf{x}^{\mathbf{t}}$, $\Delta(\mathbf{y}, \mathbf{m})$ see § 3.)

In particular, the estimate is valid if we fix $\mathbf{t} = \mathbf{0}$ and $\mathbf{x} = \mathbf{0}$:

$$\left(\frac{\gamma_0}{\mu}\right)^{\mu-M} |\Delta(\mathbf{y}, \mathbf{m})| \leq \frac{1.1}{\sqrt{2\pi\mu}} \left(\frac{e\gamma_0}{\mu}\right)^{\mu}.$$

Proof. Replacing x_j, y_j by $|x_j|, |y_j|$, we can assume that $\mathbf{x}, \mathbf{y} \geq \mathbf{0}$. Considering the sum over $\mathbf{n}$ such that $\|\mathbf{n}\| = \|\mathbf{m}\|$ (with $\mathbf{m}$ among the summands), we get

$$\mathcal{D} \leq \frac{1}{M!} \sum_{\mathbf{n}, \mathbf{t}} \frac{M!}{\mathbf{n}! \mathbf{t}! \mathbf{1}!} \mathbf{x}^{\mathbf{t}} (\mathbf{y} + \mathbf{m})^{\mathbf{n}} = \frac{1}{M!} (\|\mathbf{x}\| + \|\mathbf{y}\| + \|\mathbf{m}\|)^M,$$

which yields the first inequality (8.4).

To prove the second we must show that

$$f(\mu) = \frac{1}{2} \ln(2\pi\mu) + M \ln(\mu) - \mu - \ln(M!) < \ln(1.1). \quad (8.5)$$

For $M = 0$ we easily see that the worst value of μ is 1, and (8.5) holds. Differentiating $f(\mu)$, we obtain that the maximum for $M \geq 1$ is achieved at $\mu = M + 0.5$. Then

$$\begin{aligned} f\left(M + \frac{1}{2}\right) &= \frac{1}{2} \ln(2\pi M) + M \ln(M) - M - \ln(M!) + f_1(M), \\ f_1(M) &= \left(M + \frac{1}{2}\right) \ln\left(1 + \frac{1}{2M}\right) - \frac{1}{2}. \end{aligned}$$

Since the function $f_1(M)$ decreases, it assumes its minimum at $M = 1$. Moreover, Stirling's inequality

$$x^x e^{-x} (2\pi x)^{1/2} < (x)! < x^x e^{-x} (2\pi x)^{1/2} e^{1/(12x)}, \quad x \in \frac{1}{2}\mathbb{Z}, \quad x > 0,$$

implies that it suffices to verify (8.5) for $M = 1, 2$. A direct calculation establishes the desired result.

§ 9. The general form of auxiliary functions

We will write the auxiliary functions in modern notation, which is suitable for applying the differential operators ∂_j (introduced in (3.1)) and the zero estimates (see § 19):

$$\mathcal{F}_s(\bar{Y}') = \mathcal{F}_s(Y_0, \dots, Y_n) = \sum^{(s)} P_s(Y_0, \bar{l}) \bar{Y}^{\bar{l}}, \quad s = 0, 1, \dots, S. \quad (9.1)$$

Here the sum is taken over the elements $\bar{l}$ of some set $\mathcal{L}_s$ of multipowers, and $P_s(Y_0, \bar{l})$ are polynomials in Y_0 . We write them in terms of the basis of generalized binomial polynomials:

$$P_s(z, \bar{l}) = \sum_{l_0=0}^{L_0} p(\bar{l}') \Delta(2^{S-s} z, l_0, H)$$

with indeterminate coefficients $p(\bar{l}') = p(l_0, \dots, l_n) \in \mathbb{K}$ and some H . Then

$$\deg_{Y_0} \mathcal{F}_s(Y_0, \dots, Y_n) \leq L_0.$$

As s increases, we rarefy the sets of multipowers so that

$$\mathcal{L}_S \subseteq 2^{-1} \mathcal{L}_{S-1} \subseteq \dots \subseteq 2^{1-S} \mathcal{L}_1 \subseteq 2^{-S} \mathcal{L}_0. \quad (9.2)$$

This enables us to avoid renumbering the $p(\bar{l}')$ for different s . The structure of the sets $\mathcal{L}_s$ is described in more detail in § 10.

The notation of [1] is somewhat different. It is related to (9.1) by

$$F_s(x, \tilde{m}') = \tilde{\partial}'^{\tilde{m}'} \mathcal{F}_s(x, \alpha_1^x, \dots, \alpha_n^x) = \sum^{(s)} P_s^{(m_0)}(x, \bar{l}) \tilde{\chi}^{\tilde{m}}(\bar{l}) \bar{\alpha}^{\bar{l}x}. \quad (9.3)$$

The motivation for considering such functions and operators is as follows. Assuming that the linear form in logarithms is small and making in $\mathcal{F}_s(z, \alpha_1^z, \dots, \alpha_n^z)$, $z \in \mathbb{C}$, a substitution with small correction term

$$\alpha_n^{l_n z} \approx \exp(-(\tilde{b} \cdot \ln \bar{\alpha}) z l_n / b_n), \quad (9.4)$$

we get an analytic function

$$\mathcal{G}_s(z) = \sum^{(s)} P_s(z, \bar{l}) \prod_{j=1}^{n-1} \alpha_j^{(l_j b_n - l_n b_j) z / b_n} = \sum^{(s)} P_s(z, \bar{l}) \tilde{\alpha}^{\tilde{\chi}(\bar{l}) z / b_n}.$$

Differentiating this, we encounter the expressions above. The components $P_s(x, \bar{l})$, $\tilde{\chi}^{\tilde{m}}(\bar{l})$ and $\bar{\alpha}^{\bar{l}x}$ in (9.3) are respectively called the *polynomial part*, the *differential part* and the *exponential part*.

The proof mainly consists in constructing a non-trivial set of functions satisfying

$$F_s(x, \tilde{m}') = 0, \quad x \in \mathcal{X}_s, \quad \|\tilde{m}'\| \leq M_s, \quad s = 0, 1, \dots, S, \quad (9.5)$$

for some numbers M_s (multiplicities) and finite sets $\mathcal{X}_s \subseteq \mathbb{Z}$ (of zeros) that will be defined in the following sections. We mention some transformations applicable to these functions. The exponents l_j ($1 \leq j \leq n$) may be negative. Multiplying $\mathcal{F}$ by an appropriate power $\bar{Y}^{\bar{l}}$, we can shift all exponents to the positive domain. This fact enables us to convert $\mathcal{F}$ into a polynomial, and this will be used at the end of the proof. The powers of Y_j are determined by the changes in l_j . Multiplication by $\bar{Y}^{\bar{l}}$ does not influence the system of equations (9.5).

The novelty of the proof is that the exponents l_j may assume fractional values. More precisely, they are of the form

$$\bar{l} = \bar{v}_s + \bar{\lambda} \in \mathcal{L}_s, \quad \bar{\lambda} \in \frac{1}{N} \mathbb{Z}^n. \quad (9.6)$$

Then we get integer exponents after shifting by $\bar{v}_s$ and changing the variables by $\bar{Z} = \bar{Y}^{1/N}$. This will also be used at the end of the proof to pass to polynomials.

Under this substitution, the degree may increase by a factor of N . We also note that analytic properties of the functions are independent of the arithmetic nature of the vectors $\bar{v}_s$, and the corresponding factor can be factored out, which leaves only well-defined algebraic expressions in the brackets.

A linear change of the variable x enables us to transform the zeros (9.5) (which form an arithmetic progression) to the form $x = 0, \dots, X'$, and this will also be used at the end of the proof to apply the zero estimate. All these transformations change somewhat the coefficients of the functions but produce no significant changes in the system (9.5).

To solve (9.5), we transform it into a more convenient form. We replace the differential part $\tilde{\chi}(\bar{l})^{\tilde{m}}$ of the function F_s in (9.3) by $\Delta(2^s N \tilde{\chi}(\bar{l} - \bar{w}_s), \tilde{m})$ (which will also be called the *differential part*), where the vector $\bar{w}_s$ is determined by the relations

$$\bar{l} - \bar{w}_s \in (2^s N)^{-1} \mathbb{Z}^n, \quad \bar{l} \in \mathcal{L}_s, \quad |w_{s,j}| \leq 2^{-1-s} N^{-1}, \quad 1 \leq j \leq n. \quad (9.7)$$

Then we multiply F_s by $\frac{1}{m_0!} \bar{\alpha}^{-\bar{v}_s x} d_0(m_0) 2^{-(S-s)m_0}$ (with d_0 as in Lemma 8.1) and get

$$\begin{aligned} \Phi_s(x, \tilde{m}') &= \sum_{l_0=0}^{(s)} \sum_{l_0=0}^{L_0} a_s(\bar{l}'; x, \tilde{m}') p(\bar{l}'), \\ a_s(\bar{l}'; x, \tilde{m}') &= d_0(m_0) 2^{-(S-s)m_0} \frac{1}{m_0!} \Delta^{(m_0)}(2^{S-s} x, l_0, H) \\ &\quad \times \Delta(2^s N \tilde{\chi}(\bar{l} - \bar{w}_s), \tilde{m}) \bar{\alpha}^{\bar{\lambda} x}. \end{aligned} \quad (9.8)$$

By Lemma 3.1, the system of equations (9.5) is equivalent (for each s) to the corresponding system

$$\Phi_s(x, \tilde{m}') = 0, \quad x \in \mathcal{X}_s, \quad \|\tilde{m}'\| \leq M_s, \quad s = 0, 1, \dots, S. \quad (9.9)$$

After the substitution (9.4), the functions $\mathcal{G}_s(z)$ are replaced by the functions

$$\begin{aligned} \Psi_s(z, \tilde{m}') &= \sum_{l_0=0}^{(s)} \sum_{l_0=0}^{L_0} p(\bar{l}') 2^{-(S-s)m_0} \frac{1}{m_0!} \Delta^{(m_0)}(2^{S-s} x, l_0, H) \\ &\quad \times d_0(m_0) \Delta(2^s N \tilde{\chi}(\bar{l} - \bar{w}_s), \tilde{m}) \tilde{\alpha}^{\tilde{\chi}(\bar{\lambda})z/b_n}. \end{aligned} \quad (9.10)$$

§ 10. Description of the set of multipowers

The main tool for removing the factor n^n from the estimates of linear forms in [1] (in the strongly independent case) was a more careful choice of the set $\mathcal{L}_0$ of multipowers. The same tool was used in a simpler situation in [5]. The removal of the second factor n^n in the present paper is based on a further improvement of the construction of multipowers. We begin with the construction of the set $\mathcal{L}_0$.

Let the vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ in the extended logarithmic space $\mathcal{E}_\varkappa$ correspond to the numbers $\alpha_1, \dots, \alpha_n$ with chosen values of the logarithms (the coordinates of vectors are indexed by $\sigma \geq 1$). Then the vector $l_1 \mathbf{a}_1 + \dots + l_n \mathbf{a}_n$ in $\mathcal{E}_\varkappa$ corresponds to the number $\bar{\alpha}^{\bar{l}}$. In $\mathcal{E}_\varkappa$ we consider the lattices

$$M'_n = \langle \mathbf{a}_1, \dots, \mathbf{a}_n \rangle_{\mathbb{Z}}, \quad M_n = \langle \mathbf{a}_1, \dots, \mathbf{a}_n \rangle_{\mathbb{R}} \cap M_\varkappa(\mathbb{K}).$$

The lattice M'_n corresponds to the lattice $\mathcal{M}'_n = \mathbb{Z}^n$ in the space of multipowers $\bar{l}$. This gives the usual choice of the set $\mathcal{L}_0$ in previous papers. But the lattice M_n may contain other points of the extended logarithmic space. In other words, the condition $\bar{\alpha}^{\bar{l}} \in \mathbb{K}$ can hold not only for integer but also for fractional multipowers. It is this fact that enables us to treat the general case without losing accuracy. In the space $\mathbb{R}^n$ of multipowers we consider the lattice $\mathcal{M}_n$ that corresponds to M_n . We put

$$N = [M_n : M'_n] = [\mathcal{M}_n : \mathcal{M}'_n].$$

Then

$$\text{Vol}_n \mathcal{M}_n = \frac{1}{N} \text{Vol}_n \mathcal{M}'_n = \frac{1}{N}.$$

We describe the structure of the lattice $\mathcal{M}_n$, which is also used in the description of the remaining sets $\mathcal{L}_s$. Choose a basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ in M_n , and let $\mathcal{A}$ be the matrix with rows formed by the (integer) coordinates of the vectors $\mathbf{a}_i$ in this basis. Then $N = |\det \mathcal{A}|$, and the vectors $\mathbf{e}_j$ are expressed in terms of $\mathbf{a}_i$ using the inverse matrix $\mathcal{A}^{-1}$. The rows $\bar{u}_j$ of the matrix $\mathcal{A}^{-1}$ consist of the coordinates of the vectors $\mathbf{e}_j$ in the basis $\{\mathbf{a}_i\}$. Hence, if a vector $\mathbf{v} \in M_n$ has coordinates $\bar{\mu} = (\mu_1, \dots, \mu_n) \in \mathbb{Z}^n$ in the basis $\{\mathbf{e}_j\}$, then its coordinates in the basis $\{\mathbf{a}_i\}$ are equal to

$$\bar{\lambda} = \bar{\lambda}(\bar{\mu}) = (\lambda_1, \dots, \lambda_n) = \mu_1 \bar{u}_1 + \dots + \mu_n \bar{u}_n = \bar{\mu} \mathcal{A}^{-1}. \quad (10.1)$$

The definition of N implies that $N \bar{\lambda}(\bar{\mu}) \in \mathbb{Z}^n$.

We now define a domain containing the vectors under study. Take some $L > 0$ and define a solid $\mathcal{W}$ by the conditions

$$|w_j| \leq \frac{L}{2A_j}, \quad 1 \leq j \leq n. \quad (10.2)$$

The individual choice of bounds for each coordinate enabled Baker [14] to include the dependence on Ω in the estimate of a linear form. Taking into account the notation for A_j , we note that

$$\sup\{|\bar{w} \cdot \ln \bar{\alpha}| : \bar{w} \in \mathcal{W}\} \leq \eta_0^{-1} L, \quad \eta_0 = 2/n.$$

The removal of the factor n^n in [1] was mainly done by a forced decrease in the bound for $|\bar{w} \cdot \ln \bar{\alpha}|$. In the present paper we put

$$|\bar{w} \cdot \ln \bar{\alpha}| \leq \eta^{-1} L, \quad \eta = e^{1+(n+0.53)/\varkappa} \quad (10.3)$$

and define a solid $\mathcal{W}_0$ by the conditions (10.2), (10.3). It was shown in [1], Lemma 5.1 that

$$\text{Vol}_n \mathcal{W}_0 \geq \left(\frac{\eta}{\eta_0}\right)^{-\varkappa} \text{Vol}_n \mathcal{W} = \left(\frac{1}{2}n\eta\right)^{-\varkappa} \frac{1}{\Omega} L^n.$$

In [1] we replaced $\varkappa$ in the exponent by the more precise parameter

$$\rho = \text{rank}_{\mathbb{C}}\{\ln \alpha_1, \dots, \ln \alpha_n\} \leq \varkappa,$$

which is not taken into account in this paper. (For $\varkappa = 1$ we have $\rho = 1$ since the logarithms are assumed to be real.)

A small problem is caused by the fact that the number of integer points does not necessarily correspond to the volume of a solid. We use Blichfeldt's theorem (see [7], Ch. 3, Theorem 1) to the effect that there is a vector $\bar{v}_0$ such that

$$|\mathcal{W}_0 \cap (\mathcal{M}_n + \bar{v}_0)| > \frac{\text{Vol}_n \mathcal{W}_0}{\text{Vol}_n \mathcal{M}_n} \geq \left(\frac{1}{2}n\eta\right)^{-\varkappa} \frac{N}{\Omega} L^n.$$

We take these points for the set $\mathcal{L}_0$. Removing redundant points if necessary, we may assume that

$$|\mathcal{L}_0| = \left\lceil \left(\frac{1}{2}n\eta\right)^{-\varkappa} \frac{N}{\Omega} L^n \right\rceil + 1. \quad (10.4)$$

Moreover, we can assume that $\bar{v}_0 \in \mathcal{L}_0 \subseteq \mathcal{W}_0$. We now put

$$\begin{aligned} \bar{v}_s &= 2^{-s}\bar{v}_0, \quad s \geq 0, \\ \mathcal{L}_s &= 2^{-s}\mathcal{L}_0 \cap (\mathcal{M}_n + \bar{v}_s), \quad 0 \leq s \leq S, \end{aligned} \quad (10.5)$$

and note that this yields $\mathcal{L}_0$ for $s = 0$. Moreover, $\bar{v}_s \in \mathcal{L}_s \cap 2^{-s}\mathcal{W}_0$, and points of $\mathcal{L}_s$ are comparable modulo the lattice $\mathcal{M}_n$. Furthermore, condition (9.2) holds because

$$2\mathcal{L}_{s+1} \subseteq 2^{-s}\mathcal{L}_0 \cap (2\mathcal{M}_n + 2\bar{v}_{s+1}) \subseteq 2^{-s}\mathcal{L}_0 \cap (\mathcal{M}_n + \bar{v}_s) = \mathcal{L}_s.$$

The sets constructed in (10.5) are consistent with conditions (9.6). We note that in this paper we work with shifted sets which are contained in a centrally symmetric set, preferring this to arithmetic properties. This is convenient and enables us to simplify calculations. When studying arithmetic expressions, we factor out the number $\bar{\alpha}^{\bar{v}}$.

§ 11. The scheme of arithmetic-analytic continuation

In methods that go back to papers of Gel'fond, the proofs are based on constructing an auxiliary function with sufficiently many zeros, counting multiplicities. (This is done by an estimate usually called Siegel's lemma; see §§ 13–15.) Then one uses interpolation techniques and the assumption that the linear form is small to change the relation between zeros and their multiplicities (see §§ 16–18) in such a way that

their number becomes inadmissible (the theorem on the zero estimate, § 19). This contradicts our assumption and proves the estimate of the linear form.

In Baker's scheme the number of zeros increases at the expense of decreasing the multiplicity. On the other hand, increasing the number of zeros worsens the estimate of the linear form and thus requires maximal economy. An important novelty in [15] was the use of Kummer theory and the principle of dividing the argument to split the auxiliary function into pieces, which is equivalent to decreasing its degree. This enables one to apply the zero estimate and decrease the number of iterations. In [4], [1] and the present paper, the degree is decreased by a factor of two at each step. This gives the most efficient way of dividing the argument.

We organize the arithmetic-analytic continuation by induction on s , the number of divisions of the argument. The base of the induction ($s = 0$) will be constructed in § 15. The inductive step consists in applying the principle of dividing the argument (see § 18) to obtain equations (9.5) until the degree of the auxiliary function is as small as required.

Before dividing the argument, we must increase the number of zeros by a factor of approximately 2^n (again with some loss of multiplicity). In [1] we suggested an efficient but technically difficult scheme where the multiplicity smoothly decreases as x (the number of interpolation points) increases. In this paper we simplify this step of the proof, which worsens the estimate by a factor of order e^n and worsens the absolute constant. The preliminary reproduction of zeros is divided into n steps and proceeds by induction on $\nu = 0, \dots, n$ for a fixed s . For the base ($\nu = 0$) we take zeros with multiplicities obtained at the current step in s . At each step, the number of zeros is doubled and the multiplicity is lowered by T_s , and we also lose multiplicity T_s when dividing the argument. This scheme enables us to get constants corresponding to the result of [3].

Let $\varepsilon_0 > 0$ be some constant, and let $C_0\xi\omega$ be some (not small) expression to be defined in the next section. We put

$$\begin{aligned} c_1 &= 2(n+1)(1+\varepsilon_0), & M_0 &= [C_0\xi\omega], & T_s &= \left\lceil \frac{2^{-s}C_0\xi\omega}{c_1} \right\rceil, \\ M_{s,\nu} &= M_s - \nu T_s, & \nu &= 0, \dots, n, \\ M_{s+1} &= M_s - (n+1)T_s, & s &= 0, \dots, S. \end{aligned} \quad (11.1)$$

We estimate the multiplicity $M_{S,n}$:

$$\begin{aligned} M_{S,n} &= M_0 - \sum_{s=0}^S (n+1)T_s + T_S \\ &\geq C_0\xi\omega + T_S - 1 - \sum_s^* 2^{-s} \frac{C_0\xi\omega}{2(1+\varepsilon_0)} \geq \frac{\varepsilon_0 C_0\xi\omega}{1+\varepsilon_0}. \end{aligned} \quad (11.2)$$

(The last sum $\sum^{(*)}$ consists of those terms in s that exceed $n+1$. Hence 1 may be replaced by the remainder of the geometric progression.) The parameter ε_0 enables us to control the remaining multiplicity.

We now describe the sets of interpolation points. Choose some (large) $G > 0$ and put

$$X_s = \left\lceil \frac{2DG}{T_s + 1} \right\rceil, \quad s \geq 0 \quad (11.3)$$

($\lceil x \rceil$ means the least integer $\geq x$). Finally, we define the sets $\mathcal{X}_{s,\nu}$ for (9.5) by the conditions

$$\begin{aligned} \mathcal{X}_{0,0} &= \mathcal{X}_0 = \{0, \pm 1, \dots, \pm X_0\}, \\ \mathcal{X}_{s,0} &= \mathcal{X}_s = \{\pm 1, \pm 3, \dots, \pm(2X_s - 1)\}, \quad s > 0, \\ \mathcal{X}_{s,\nu} &= \{0, \pm 1, \dots, \pm 2^\nu X_s\}, \quad 1 \leq \nu \leq n, \quad s \geq 0. \end{aligned} \quad (11.4)$$

These definitions of T_s and $\mathcal{X}_{s,\nu}$ guarantee that the following two-sided inequality holds:

$$2^{\nu+2}DG \leq |\mathcal{X}_{s,\nu}|(T_s + 1) \leq 2^\nu(4DG + 3T_s + 3). \quad (11.5)$$

Thus we shall prove equations that include conditions (9.5) for $\nu = 0$:

$$F_s(x, \tilde{m}') = 0, \quad x \in \mathcal{X}_{s,\nu}, \quad \|\tilde{m}'\| \leq M_{s,\nu}. \quad (11.6)$$

We do this by double induction, passing from (s, ν) to $(s, \nu + 1)$ for $\nu < n$ (see § 17) and from (s, n) to $(s + 1, 0)$ (see § 18), each time losing multiplicity T_s . At the end of the proof we need only conditions (11.6) for $(s, \nu) = (S, n)$.

We note that it was the accurate organization of the arithmetic-analytic continuation that enabled us to remove the individual lower bounds for A_j in [1]. Although everything is given in the notation, the reasoning contains some delicate points. Here we present more explicit comments.

As already noted, we can transform the auxiliary function to a polynomial of degrees NL/A_j in the variables Y_j . Division by A_j usually decreases the degree and simplifies the reasoning. But if the A_j are small, the degree increases. The new factor N has the same disadvantage. This prevents us from applying the zero estimates. One cannot overcome this difficulty in the usual scheme of arithmetic-analytic continuation, but division of the argument enables us to remove this redundant factor by making several additional inductive steps in s , where each step decreases the degree by a factor of 2.

There is another delicate point in the reasoning. The degree of the interpolation polynomial $|\mathcal{X}_{s,\nu}|(T_s + 1)$ plays a key role in the proof and in the resulting estimate (see § 16). We see from (11.3) that this degree is almost unaltered at every step, and (11.1) shows that if $T_s \gg 1$ (which is just the case in previous papers for all s), then the multiplicity decreases by a factor of approximately two, and hence the number of points doubles. But now the case $T_s = 0$ becomes possible. It requires another approach: as soon as we get $T_s = 0$, we do not change $|\mathcal{X}_{s,\nu}|$ (increasing of $|\mathcal{X}_{s,\nu}|$ would sharply worsen the estimate of the linear form in logarithms), and the degree of the interpolation polynomial is accordingly preserved. This is expressed by the formulae (11.3), (11.4).

§ 12. Choosing the parameters

In this section we explicitly indicate the values of the parameters that are necessary in the proof of the main theorem but are absent from its statement. The meaning of these parameters has been explained in previous sections. For completeness and brevity we recall some earlier formulae. We also give preliminary estimates of the expressions mentioned.

As already noted, we may assume that $n \geq 2$. In the extended logarithmic space we consider linearly independent vectors $\mathbf{v}_j = \mathbf{v}(\alpha_j)$ corresponding to the chosen values of $\ln \alpha_j$. In Proposition 5.1 we introduced the expressions

$$M_n = M_{\mathcal{X}}(\mathbb{K}) \cap \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{R}}, \quad M'_n = \langle \mathbf{v}_1, \dots, \mathbf{v}_n \rangle_{\mathbb{Z}}, \quad N = [M_n : M'_n],$$

which satisfy the inequalities (5.1), (5.2). We introduce the parameters

$$\eta = e^{1+(n+0.53)/\kappa}, \quad \xi = \frac{8}{n!}(n+2.1)(2(n+1))^{n+1} \left(\frac{1}{2}n\eta \right)^{\kappa},$$

$$\varepsilon_0 = 3n^3/\xi, \quad \varepsilon_1 = \xi_0(n)/(2\xi), \quad \varepsilon_2 = (4(n+1)(2.25n + 1.5 \ln n + 13))^{-1}.$$

Put

$$C_0 = \ln(e^{4.4n+7} n^{5.5} N D^2 \ln(eD)), \quad W_0 = \ln(1.5eBND \ln(eD)).$$

(This differs from (2.2) by the presence of N , so that C_0, W_0 in (2.2) are obtained by putting $N = 1$.) Furthermore, we put

$$\gamma = \frac{C_0(1)W_0(1)}{C_0(N)W_0(N)}N, \quad \omega = \frac{\gamma}{N}D\Omega = \frac{\gamma}{N}DA_1 \dots A_n.$$

Note that this choice of γ yields

$$C_0(N)W_0(N)\omega(N, \gamma) = C_0(1)W_0(1)\omega(1, 1),$$

and we can simply write $C_0W_0\omega$, removing the parameters N, γ from estimates of linear forms. In what follows we thus assume that C_0, W_0 depend on N as defined in this section, but we usually omit N from the notation C_0, W_0 .

Suppose that

$$S = \lceil \log_2(0.5e^{-3}n^2\xi N D \ln(eD)) \rceil, \quad G = C_0W_0\xi\omega, \quad c_1 = 2(n+1)(1+\varepsilon_0)$$

(see (11.1)),

$$M_0 = [C_0\xi\omega], \quad T_s = \left\lceil \frac{1}{c_1}2^{-s}C_0\xi\omega \right\rceil, \quad X_s = \left\lceil \frac{2DG}{T_s+1} \right\rceil, \quad s \geq 0$$

(see (11.3)),

$$L = \frac{1}{c_1}C_0\xi\omega, \quad L_0 = [W_0\xi\omega], \quad H = \lceil 0.96 \ln(1.5BND \ln(eD)) \rceil$$

(see § 8), and let the parameters K, c in Lemmas 8.1, 8.2 be equal to 2^{S-s} .

We present simple estimates of these parameters. As N increases, the expression $C_0(N)W_0(N)$ obviously increases, and $C_0(N)W_0(N)/N$ decreases. Hence we have $1 \leq \gamma \leq N$. By (5.1), the choice of ε_1 yields that

$$\varepsilon_1 \xi \omega \min\{C_0, W_0\} \geq 1.$$

By (6.2), we may assume that $W_0 > C_0/2$, and ε_2 satisfies $2c_1 W_0 \varepsilon_2 > 1$. Thus $\varepsilon_1, \varepsilon_2$ estimate the relative error in replacing M_0, L_0, X_0 by the right-hand sides of the corresponding formulae without rounding off.

We also give a preliminary estimation of the parameters, which will be used in what follows to simplify expressions. As n grows, we have

$$\begin{aligned} \xi &= (2e^2)^n n^{\varkappa+1.5} O(1) > 1.2 \cdot 10^5, & M_0 &\geq 2\xi/\xi_0(n) > 1.3 \cdot 10^4, \\ NL/A_j &\geq 2\xi/(2(n+1)\xi_0(n-1)) > 2.8 \cdot 10^4, \\ \varepsilon_0 &= (2e^2)^{-n} n^{1.5-\varkappa} O(1) < 2 \cdot 10^{-4}, \\ \varepsilon_1 &= (2e^2/3.1)^{-n} n^{-1.5} O(1) < 10^{-4}, \\ \varepsilon_2 &= O(n^{-2}) < 5 \cdot 10^{-3}. \end{aligned}$$

Thus the parameters $\varepsilon_0, \varepsilon_1$ are exponentially small, and $M_0, L_0, L, NL/A_j$ increase exponentially with n . The parameter ε_2 is also small. The exponential growth of $M_0, L_0, L, NL/A_j$ worsens the estimate of the linear form but simplifies the proof since we can take small values of the parameters ε_i . We note at once that the contribution of ε_i in all subsequent estimates will increase the factors by at most 0.1. For example, we shall use the following rounding off:

$$\begin{aligned} 2(n+2) \left(\frac{2n+1}{\varkappa} + 2 \right) &> e^{0.53} (n+2.1) \left(\frac{2n+1}{\varkappa} + 2 + \frac{1}{\eta} + n\varepsilon_1 \right) (1+2\varepsilon_2), \\ n+2.1 &> (1+2\varepsilon_2) \left((1+\varepsilon_0)(1+n\varepsilon_1) \right)^{n+1} (n+2+2n\varepsilon_2+3n\varepsilon_1). \end{aligned} \quad (12.1)$$

This can be verified directly for n small and remains valid for n large because of the asymptotic behaviour of ε_i .

At the inductive step (s, ν) , the complex variable z ranges over the set $|z| \leq R_{s,\nu}$, where

$$R_{s,\nu} = 2^s \eta |\mathcal{X}_{s,\nu}| (T_s + 1) / L.$$

We assume that the linear form satisfies

$$\ln |\Lambda| < -2^{n+1} \left(\frac{2n+1}{\varkappa} + 2 + \frac{1}{\eta} + n\varepsilon_1 \right) \xi D \omega C_0 W_0. \quad (12.2)$$

By (12.1), this estimate is contrary to (2.3). Hence, to prove Theorem 2.1 it suffices to get a contradiction from (12.2).

We illustrate the estimation of various expressions by the example of an upper bound for M_0 . We split the formula for M_0 into two components: the part bounded by $\exp(aC_0)$ with the corresponding a on account of the choice of C_0 , and the

remaining part x , to which we apply the inequality $x \leq e^{x/e}$. On account of (5.1) and the formula for ε_1 we finally have

$$M_0 + n \leq ((1 + n\varepsilon_1)\xi)(C_0\omega) \leq e^{0.7C_0} e^{C_0\omega/e} \leq e^{0.7C_0\varepsilon_1\xi\omega W_0} e^{C_0\omega/e} \leq e^{\varepsilon_1 G}.$$

We analogously get upper estimates for L_0 , L , L/A_j , G , X_0 , 2^S and other similar expressions.

Furthermore, we may assume that $\mathbb{K} = \mathbb{Q}(\alpha_1, \dots, \alpha_n)$, since increasing the degree the field worsens the estimate of the linear form.

§ 13. Siegel's lemma

In the theory of numbers, results stating that an incomplete system of linear inhomogeneous equations has a non-zero solution with comparatively small values of the variables, are usually called Siegel lemmas. For algebraic linear equations, a modern estimate of solutions was obtained in [15]. We use a sharpened version [8] which takes the weight coefficients into account. We begin by introducing some notation.

To study vectors and linear forms over $\mathbb{K}$, the corresponding (Euclidean) lengths were introduced in [8]. Suppose that we have a set of positive numbers (weight coefficients)

$$\mathcal{Q} = \{q_{j,\sigma} : 1 \leq j \leq J; \sigma \geq 1\}.$$

We assume that $q_{j,\sigma} = q_{j,\tau}$ for complex-conjugate σ, τ , and that for $\sigma > D$ we have $q_{j,\sigma} = (\text{Norm } \mathfrak{p}_\sigma)^{l_{j,\sigma}}$ with some $l_{j,\sigma} \in \mathbb{Z}$. Moreover, $q_{j,\sigma} = 1$ for almost all σ . We denote by $\mathcal{Q}'$ the set of reciprocals of the members of $\mathcal{Q}$.

We define the *local* and *absolute lengths* of a vector

$$\mathbf{a} = (a_1, \dots, a_J) \in \mathbb{K}^J$$

with weight coefficients $\mathcal{Q}'$ by the equations

$$\begin{aligned} \|\mathbf{a}\|_{\mathcal{Q}'}^{(\sigma)} &= \left(\sum_{j=1}^J |a_j|_\sigma^2 q_{j,\sigma}^2 \right)^{1/2}, \quad 1 \leq \sigma \leq D, \\ \|\mathbf{a}\|_{\mathcal{Q}'}^{(\sigma)} &= \max\{|a_j|_\sigma q_{j,\sigma} : 1 \leq j \leq J\}, \quad \sigma > D, \\ \|\mathbf{a}\|_{\mathcal{Q}'} &= \left(\prod_{\sigma=1}^\infty \|\mathbf{a}\|_{\mathcal{Q}'}^{(\sigma)} \right)^{1/D} \end{aligned}$$

(we include the subscript $\mathcal{Q}'$ to make the dependence of the norm on the weight coefficients consistent with (4.3) and the notation of [8]). We fix the notation $\|\cdot\|_{\mathcal{Q}'}$ for these norms.

Lemma 13.1 ([8], Theorem 3). *Given any vectors $\mathbf{a}_i \in \mathbb{K}^J$, $1 \leq i \leq I$, and a set $\mathcal{Q}$ of weight coefficients, we put*

$$q_j = \left(\prod_{\sigma=1}^\infty q_{j,\sigma} \right)^{1/D}, \quad 1 \leq j \leq J, \quad q = (q_1 \dots q_J)^{1/J}.$$

Then the system of linear equations

$$\mathbf{a}_i \cdot \mathbf{p} = 0, \quad 1 \leq i \leq I$$

of rank $r \leq I < J$ has a non-zero solution $\mathbf{p} \in \mathbb{K}^J$ such that

$$\|\mathbf{p}\|_{\mathcal{Q}} \leq \gamma \left(q^{-J} \prod_{i=1}^I \max\{q, \|\mathbf{a}_i\|_{\mathcal{Q}'}\} \right)^{1/(J-I)}, \quad (13.1)$$

where we may take $\gamma = 1$ in the cases $r = 0$ and $r = I = J - 1$, and

$$\gamma = 2\rho_{(J-I)D} |\Delta(\mathbb{K})|^{1/(2D)} / \sqrt{D}$$

in the case $r = I < J - 1$. Here ρ_m is the radius of an m -dimensional ball of volume 1, and $\Delta(\mathbb{K})$ is the discriminant of the field $\mathbb{K}$. Moreover, we may always take

$$\gamma = \sqrt{J} |\Delta(\mathbb{K})|^{1/(2D)}. \quad (13.2)$$

Remark 13.1. The prototype of (13.2) in [8] contained a coefficient 0.5 for $J > 4$. This was confirmed by an inequality based on Stirling's formula for the factorial appearing in ρ . This inequality was valid asymptotically and was numerically verified for small J . Subsequent examination has shown that we may omit 0.5 unless $J > 68$ and that (13.2) (without the factor 0.5) is valid for all J .

Remark 13.2. Although we do not need the following fact in this paper, we note that the local lengths of the solution $\mathbf{p}$ are also estimated in the proof of Siegel's lemma in [8]. This yields $\|\mathbf{p}\|_{\mathcal{Q}}^{(\sigma)} \leq 1$ for $\sigma > D$. If $q_{j,\sigma} = 1$, $\sigma > D$, it follows that $\mathbf{p}$ has algebraic integer coefficients.

The following simple lemma uses the notions introduced in §5 and serves to study the components of vectors appearing in what follows.

Lemma 13.2. *If the components of some set $\{\gamma_j\}$ are decomposed into factors $\gamma_j = \delta_j \varepsilon_j$ in the field $\mathbb{K}$, then*

$$\begin{aligned} \|\{\gamma_j\}\|_{\mathcal{Q}'}^{(\sigma)} &\leq \exp(h^{(\sigma)}(\{\delta_j\})) \|\{\varepsilon_j\}\|_{\mathcal{Q}'}^{(\sigma)}, \quad \sigma \geq 1, \\ \|\{\gamma_j\}\|_{\mathcal{Q}'} &\leq \exp(h(\{\delta_j\})) \|\{\varepsilon_j\}\|_{\mathcal{Q}'} \end{aligned}$$

Proof. The first inequality means that the maximal element corresponding to $h^{(\sigma)}(\{\delta_j\})$ may be factored out from the norms under consideration. The second is obtained by taking the product of the first over σ .

This lemma enables us to study the norms of vectors by splitting them into components (the number of factors may be two or more). We also recall that $\mathbf{a}, \mathbf{p} \in \mathbb{K}^J$ satisfy

$$|\mathbf{a} \cdot \mathbf{p}|_{\sigma} \leq \|\mathbf{a}\|_{\mathcal{Q}'}^{(\sigma)} \cdot \|\mathbf{p}\|_{\mathcal{Q}}^{(\sigma)}. \quad (13.3)$$

§ 14. The discriminant

To estimate the discriminant $\Delta(\mathbb{K})$ appearing in Siegel's lemma, we need the following assertion.

Lemma 14.1 ([1], Lemma 6.2). *Suppose that $\alpha_1, \dots, \alpha_n$ are algebraic, $\mathbb{K} = \mathbb{Q}(\alpha_1, \dots, \alpha_n)$, $D = [\mathbb{K} : \mathbb{Q}]$. Then*

$$\ln |\Delta(\mathbb{K})| \leq D \ln D + 2D(D-1) \max\{h(\alpha_j) : 1 \leq j \leq n\}.$$

We note that the results of [1] were presented at conferences in Hungary (Eger, 29.07–2.08.1996) and Byelorussia (Minsk, 1–8.09.1996). This lemma is stated in [17] with reference to Masser's letter to Wüstholz dated 08.12.1990.

The parameter N is absent from the estimate of $|\Delta(\mathbb{K})|$. This prevents us from including N in the statement of the theorem. Moreover, this creates a difficulty in the proof, which will be avoided in this section and motivates the introduction of the parameter γ in § 12.

Lemma 14.2. *In the above notation we have*

$$\frac{1}{2D} \ln |\Delta(\mathbb{K})| \leq 0.3\varepsilon_1 G.$$

Proof. Using (5.1), (5.2), the estimates for C_0 , W_0 and the definition of ε_1 , we have

$$\begin{aligned} 0.3\varepsilon_1 G &= 0.3\varepsilon_1 \xi C_0(N) W_0(N) \omega(N, \gamma) = 0.3\varepsilon_1 \xi C_0(1) W_0(1) \omega(1, 1) \\ &\geq (0.25 + 0.05) \frac{1}{2} 3.1^n n^{\frac{D}{\varkappa}} \Omega C_0 W_0 \geq \frac{1}{2} \ln D + \max A_j. \end{aligned}$$

The lemma now follows immediately from Lemma 14.1.

To conclude the section, we note that the interpolation determinant technique of proving estimates for linear forms in logarithms (see [18]) makes no use of the discriminant $\Delta(\mathbb{K})$ and hence suggests that one might be able to improve the estimates in this respect.

§ 15. Construction of the initial auxiliary function

The main purpose of this section is to construct (by Siegel's lemma) a non-zero set of non-large coefficients $p(\bar{l}') \in \mathbb{K}$ forming a solution of system (9.9) for $s = 0$ and to estimate this solution.

The variables $p(\bar{l}')$ of this system are labeled by multi-indices $\bar{l}' = (l_0, \bar{l}) = (l_0, l_1, \dots, l_n)$, where $0 \leq l_0 \leq L_0$ and $\bar{l} \in \mathcal{L}_0$. The number of variables for $s = 0$ is $J = (L_0 + 1)|\mathcal{L}_0|$. The equations are labeled by $(x, \tilde{m}') = (x, m_0, \dots, m_{n-1})$, where $x \in \mathcal{X}_0$, $\|\tilde{m}'\| \leq M_0$. The number of equations (9.9) for $s = 0$ equals

$$I = (2X_0 + 1) \binom{M_0 + n}{n}.$$

We write the set of coefficients $a_s(\bar{l}'; x, \tilde{m}')$ in (9.8) as a vector $\mathbf{a}_s(x, \tilde{m}')$.

We also define weight coefficients for Lemma 13.1 by

$$\begin{aligned} q_\sigma(\vec{l}') &= q(\vec{l}') = e^{C_0(L_0 - l_0)}, \quad 1 \leq \sigma \leq D, \\ q_\sigma(\vec{l}') &= 1, \quad \sigma > D. \end{aligned} \quad (15.1)$$

Then the parameter q in Lemma 13.1 is given by

$$q = \exp\left(\sum_{l_0=0}^{L_0} \sum^{(0)} C_0 \frac{L_0 - l_0}{J}\right) = \exp\left(C_0 \sum_{l_0=0}^{L_0} \frac{L_0 - l_0}{L_0 + 1}\right) = e^{C_0 L_0/2}. \quad (15.2)$$

The following preparation is also necessary for later sections. We shall apply Lemma 8.3 to

$$\mu = C_0 \xi \omega, \quad \gamma_0 = C_0 \xi \omega B N 1.5 D \ln(eD), \quad (15.3)$$

and to the vectors $\mathbf{x}, \mathbf{y}$ with coordinates

$$\begin{aligned} x_j &= \frac{1}{b_n} (b_n l_j - b_j l_n) \ln \alpha_j \in \mathbb{C}, \quad 1 \leq j \leq n-1, \\ y_j &= 2^s N (b_n (l_j - w_{s,j}) - b_j (l_n - w_{s,n})) \in \mathbb{Z}, \quad 1 \leq j \leq n-1. \end{aligned}$$

Let us estimate $\|\mathbf{x}\|$, taking into account the notation for A_j , B and conditions (10.2). We have

$$|x_j| \leq \frac{1}{|b_n|} (|b_n l_j| + |b_j l_n|) |\ln \alpha_j| \leq \frac{1}{2} L(1+B).$$

To estimate $\|\mathbf{y}\|$, we must also use (9.7):

$$\begin{aligned} |b_n 2^s N (l_j - w_{s,j})| &\leq \frac{1}{2} |b_n| \left(\frac{NL}{A_j} + 1 \right) \leq \frac{BLN}{2A_j} \left(1 + \frac{1}{NL/A_j} \right), \\ |b_j 2^s N (l_n - w_{s,n})| &\leq \frac{1}{2} |b_j| \left(\frac{NL}{A_n} + 1 \right) \leq \frac{BLN}{2A_j} \left(1 + \frac{1}{NL/A_n} \right), \end{aligned}$$

and on account of (5.3) we have

$$\|\mathbf{x}\| + \|\mathbf{y}\| \leq (n-1) BNL \left(\frac{1}{1.5} + \frac{1}{1.5B} + 1 + \max_j \frac{1}{NL/A_j} \right) 1.5 D \ln(eD).$$

Taking into account the values of B , L , NL/A_n and using the formulae for M_0 , L in § 12 and for μ , γ_0 in (15.3), we get

$$M_0 + \|\mathbf{x}\| + \|\mathbf{y}\| \leq BNC_0 \xi \omega 1.5 D \ln(eD) = \gamma_0.$$

Consider the denominator $d_0(m_0)$ defined in Lemma 8.1. Here $m_0 \leq M_0 - M$, $M = \|\tilde{m}\|$. Hence (8.2) and the definitions of H , γ_0 imply that

$$d_0(m_0) \leq e^{1.04H(\mu-M)} \leq (\gamma_0/\mu)^{\mu-M}$$

and also

$$d_0(m_0) (M + \|\mathbf{x}\| + \|\mathbf{y}\|)^M \frac{1}{M!} \leq \frac{1.1}{\sqrt{2\pi\mu}} \left(\frac{e\gamma_0}{\mu} \right)^\mu \leq 0.01 e^G, \quad (15.4)$$

which together with (8.4) enables us to get the required estimates.

Lemma 15.1. *Suppose that $s \leq S$, $\|\tilde{m}'\| \leq M_0$, $x \in 2^{s-S}\mathbb{Z}$, $|x| \leq R_{s,\nu}$. Then*

$$\|\mathbf{a}_s(x, \tilde{m}')\|_{\mathcal{Q}'} \leq 0.02 \exp\left(\frac{1}{D} 2^{-s} L n |x| + 2G + n\varepsilon_1 G\right). \quad (15.5)$$

Proof. We estimate separately the contributions of factors appearing in the components of the vectors $\mathbf{a}$.

To estimate the contribution of the exponential part $\bar{\alpha}^{\bar{\lambda}x}$, $\bar{\lambda} \in \mathcal{L}_s - \bar{v}_s \subseteq \mathcal{M}_n$, we take into account the construction of $\mathcal{L}_s$, $\mathcal{W}_0(L)$ in § 10. Using the notation of § 5 and Lemma 5.3, we have

$$h^{(\sigma)}\{\bar{\alpha}, x(\mathcal{L}_s - \bar{v}_s)\} \leq \sum_{j=1}^n \max_{\pm} \left\{ x \left(\pm \frac{2^{-s}L}{2A_j} - v_{s,j} \right) \ln |\alpha_j|_{\sigma} \right\}. \quad (15.6)$$

We take the product over all $\sigma \geq 1$ and extract the root of degree D to obtain $\|\mathbf{a}\|_{\mathcal{Q}'}$. Using definition (2.1) of A_j , we get $2^{-s}Ln|x|/D$ in the first term, and the term $v_{s,j} \ln |\alpha_j|_{\sigma}$ disappears after summing over σ by the product formula.

To analyze the differential part $\Delta(2^s N \tilde{\chi}(\bar{l} - \bar{w}_s), \tilde{m})$, we note that the first argument is integer-valued. (This is the reason why we replace the shift by $\bar{v}_s$ with the shift by $\bar{w}_s$ and multiply everything by $2^s N$.) As a result, the binomial expression is also integer-valued. (This together with the division by $m!$ explains the passage from ordinary to binomial polynomials.) Applying Lemma 8.3 and (15.4), we get

$$d_0(m_0) |\Delta(2^s N \tilde{\chi}(\bar{l} - \bar{w}_s), \tilde{m})| \leq 0.01 e^G.$$

We estimate the polynomial part together with the weight coefficient $q(\bar{l}')$ in (15.1). Using Lemma 8.2 for $\tau = 0$, the choice of C_0 and the fact that $|x| \leq R_{s,\nu}$, we get

$$\begin{aligned} & \frac{1}{m_0!} 2^{-(S-s)m_0} \left| \Delta^{(m_0)}(2^{S-s}x, l_0, H) \right| q(\bar{l}') \\ & \leq \delta(l_0, m_0, 0) \left(e \left(1 + 2^{S-s} \frac{|x|+1}{H} \right) \right)^{l_0} e^{C_0(L_0-l_0)} \\ & \leq \delta(l_0, m_0, 0) e^{C_0 L_0} \leq \delta(l_0, m_0, 0) e^G, \end{aligned}$$

where

$$\sum_{l_0=0}^{L_0} \delta(l_0, m_0, 0) \leq 1.6 e^{H/e} \leq 1.6 e^{0.4\varepsilon_1 G}.$$

It remains to estimate J . By (10.4) we have

$$\begin{aligned} J &= (L_0 + 1) \left(\left[\left(\frac{1}{2} n \eta \right)^{-\varkappa} \frac{N}{\Omega} L^n \right] + 1 \right) \\ &\leq 2(1 + \varepsilon_1) W_0 \xi \omega \left(\frac{1}{2} n \eta \right)^{-\varkappa} \frac{N}{\Omega} \left(\frac{C_0 \xi \omega}{c_1} \right)^n \\ &= \left(2(1 + \varepsilon_1) \left(\frac{1}{2} n \eta \right)^{-\varkappa} \frac{\xi}{c_1} \frac{(20n\xi/\xi_0 e)^n}{(2c_1 W_0)^{n-1}} \gamma D \right) \left(\frac{e C_0 W_0 \omega \xi_0}{20n} \right)^n \\ &\leq e^{(n-0.6)C_0} e^{0.1\varepsilon_1 G} \leq e^{(n-0.5)\varepsilon_1 G}. \end{aligned}$$

The exponent $(n - 0.6)C_0$ is obtained since γD occurs with power 1 in the bracket, and the dependence of the inequality on n is verified numerically, taking into account that $2c_1W_0 > 1/\varepsilon_2$. As a result, we get

$$J \leq e^{(n-0.5)\varepsilon_1 G}. \quad (15.7)$$

Combining these estimates and taking into account the factor $\sqrt{J}$ in the Euclidean norm, we get (15.5).

Lemma 15.2. *The system (9.9) with $s = 0$ has a non-zero solution $\mathbf{p} \in \mathbb{K}^J$ such that*

$$\|\mathbf{p}\|_{\mathcal{Q}} \leq e^{n\varepsilon_1 G}.$$

Proof. We first estimate I and J . We have

$$\begin{aligned} I &= |\mathcal{X}_0| \Delta(M_0 + 1, n) \leq (2X_0 + 1) \frac{1}{n!} (C_0(1 + n\varepsilon_1)\xi\omega)^n \\ &\leq 4c_1DW_0 \left(1 + \frac{1}{4}\varepsilon_2\right) \frac{1}{n!} (C_0\xi(1 + n\varepsilon_1)\omega)^n \end{aligned} \quad (15.8)$$

and

$$\begin{aligned} J &= (L_0 + 1)|\mathcal{L}_0| > W_0\xi\omega \left(\frac{1}{2}n\eta\right)^{-\varkappa} \frac{N}{\Omega} \left(\frac{C_0\xi\omega}{c_1}\right)^n \\ &= W_0\gamma D\xi \left(\frac{1}{2}n\eta\right)^{-\varkappa} \left(\frac{C_0\xi\omega}{c_1}\right)^n. \end{aligned} \quad (15.9)$$

We now estimate the product in (13.1). We note that the bound (15.4) for $\|\mathbf{a}_0(x, \tilde{m}')\|_{\mathcal{Q}'}$ exceeds q in (15.2), whence

$$\begin{aligned} \Pi &= \prod_{i=1}^I \max\{q, \|\mathbf{a}_i\|_{\mathcal{Q}'}\} = \prod_{x=-X_0}^{X_0} \prod_{|\tilde{m}'| \leq M_0} \|\mathbf{a}_0(x, \tilde{m}')\|_{\mathcal{Q}'} \\ &\leq \exp \left\{ InL \frac{X_0(X_0 + 1)}{(2X_0 + 1)D} + (2 + n\varepsilon_1)GI \right\} \\ &\leq \exp \left(IG \left(n + \frac{3}{2}n\varepsilon_2 + 2 + n\varepsilon_1 \right) \right). \end{aligned}$$

We claim that $q^J \geq \Pi$. Indeed, consider the ratio

$$\frac{\ln \Pi}{J \ln q} \leq \frac{IG(n + 2n\varepsilon_2 + 2 + n\varepsilon_1)}{JC_0L_0/2}.$$

Replacing I, J by their estimates in (15.8), (15.9), using the notation of § 12 and cancelling identical terms, we get

$$\frac{\frac{8}{n!}(1 + 2\varepsilon_2)(n + 2n\varepsilon_2 + 2 + n\varepsilon_1)\left(\frac{1}{2}n\eta\right)^{\varkappa}((1 + n\varepsilon_1)c_1)^{n+1}}{\xi\gamma}.$$

By the choice of ξ and (12.1) this number does not exceed 1. It follows that the expression in the brackets in (13.1) does not exceed 1. Thus the conditions of the present section hold, mainly due to the choice of ξ .

To estimate the J in (13.1) from above, we use (15.7). To estimate $|\Delta(\mathbb{K})|$, we use Lemma 14.2. The coefficient of $\varepsilon_1 G$ in the exponent of the inequality of the lemma is $(n - 0.5)/2 + 0.3 < n$. This proves Lemma 15.2.

We denote by $\bar{v}_0$ a vector in the space of multipowers corresponding to some non-zero value of $p(\bar{l}')$.

§ 16. An interpolation formula

In this section we give a quantitative version of the interpolation formula based on Cauchy's residue theorem. Such lemmas form a standard element in the technique of arithmetic-analytic continuation. The statement of the lemma is taken from [1] with a simplification: we consider only points of equal multiplicity.

Given a finite set $\mathcal{X} \subseteq \mathbb{C}$ (of interpolation points) and a non-negative integer T ($T + 1$ is the multiplicity of the points), we put

$$Q(z) = \prod_{x \in \mathcal{X}} (z - x)^{T+1}.$$

If $f(z)$ is a function analytic in a sufficiently large disc, then

$$\begin{aligned} f(z) &= f_1(z) + f_2(z), \\ f_1(z) &= \frac{1}{2\pi i} \int_{\Gamma(0, R)} \frac{Q(z)}{Q(\zeta)} \cdot \frac{f(\zeta) d\zeta}{\zeta - z}, \\ f_2(z) &= \sum_{x \in \mathcal{X}} \sum_{\tau=0}^T \frac{f^{(\tau)}(x)}{2\pi i \tau!} \int_{\Gamma(x, r)} \frac{Q(z)}{Q(\zeta)} \cdot \frac{(\zeta - x)^\tau d\zeta}{\zeta - z}. \end{aligned} \quad (16.1)$$

Here the circle $\Gamma(x, r)$ with centre x and radius r is oriented clockwise and its interior contains neither z nor other points of $\mathcal{X}$, and $\Gamma(0, R)$ is oriented anticlockwise and surrounds z and all points of $\mathcal{X}$. The expression $Q(z)/Q(\zeta)$ is called the interpolation part.

Lemma 16.1 ([1], Lemma 9.1). *Suppose that $T \geq 0$, $\mathcal{X} \subseteq \mathbb{R}$, $z \in \mathbb{R} \setminus \mathcal{X}$, the points of $\mathcal{X}$ form an arithmetic progression with common difference $\delta > 0$, and $\mathcal{X}$ is symmetric with respect to 0. Put*

$$z_0 = \max \left\{ |z|, \frac{1}{2} \delta |\mathcal{X}| \right\}, \quad r = \frac{1}{2} \min \{ \delta, \min \{ |z - x| : x \in \mathcal{X} \} \}.$$

If $\zeta \in \mathbb{C}$ and $|\zeta| > z_0$, then

$$\left| \frac{Q(z)}{Q(\zeta)} \right| \leq \left(\frac{z_0}{|\zeta|} \right)^{|\mathcal{X}|(T+1)}. \quad (16.2)$$

If $\zeta \in \Gamma(x, r)$ with $x \in \mathcal{X}$, then

$$\left| \frac{Q(z)}{Q(\zeta)} \right| \leq \left(\frac{2ez_0}{\delta |\mathcal{X}|} \right)^{|\mathcal{X}|(T+1)}. \quad (16.3)$$

§ 17. Extrapolation of zeros

The purpose of this section is to prove the inductive step in ν with a fixed s for the equations (11.6). We transform (11.6) into a form similar to (9.9):

$$\begin{aligned} \Phi_s(x, \tilde{m}') &= 0, & x \in \mathcal{X}_{s,\nu}, & \quad \|\tilde{m}'\| \leq M_{s,\nu}, \\ \nu &= 0, \dots, n, & s &= 0, 1, \dots, S, \end{aligned} \quad (17.1)$$

with $\Phi_s(x, \tilde{m}')$ defined in (9.8). Thus we assume that conditions (17.1) hold for some s and ν ($0 \leq \nu < n$) and claim that they hold for $\nu + 1$.

To prove this, we use the product formula: if we know that

$$\prod_{\sigma=1}^{\infty} |\Phi_s(z, \tilde{m}')|_{\sigma} < 1, \quad z \in \mathcal{X}_{s,\nu+1}, \quad (17.2)$$

then $\Phi_s(z, \tilde{m}') = 0$. Note that for $z \in \mathbb{Z}$ we have

$$\Phi_s(z, \tilde{m}') \in \mathbb{K}.$$

For $\sigma > \varkappa$ we trivially estimate $|\Phi_s(z, \tilde{m}')|_{\sigma}$ by (13.3), (15.5) and Lemma 15.2. For $\sigma = 1$ we apply (9.4) and pass to the functions $\Psi_s(\zeta, \tilde{m}')$ in (9.10). Assuming that the linear form is small, we estimate $\Psi_s(\zeta, \tilde{m}')$ with the help of the interpolation technique of § 16, where the set of interpolation points is $\mathcal{X} = \mathcal{X}_{s,\nu}$ with multiplicities $T = T_s$. In the complex case, the estimate for $\sigma = 2$ is the same as for $\sigma = 1$. For the function $f(\zeta)$ in (16.1) we take

$$f(\zeta) = \Psi_s(\zeta, \tilde{m}') \bar{\alpha}^{\bar{v}_s z} \prod_{\sigma > \varkappa} |\Phi_s(z, \tilde{m}')|_{\sigma}^{1/\varkappa}. \quad (17.3)$$

The use of (9.4) and the passage from the functions Φ to Ψ produce an error

$$\begin{aligned} |\bar{\alpha}^{\bar{l}\zeta} - \bar{\alpha}^{\tilde{\chi}(\bar{l})\zeta/b_n}| &= |e^{-\Lambda\zeta l_n/b_n} - 1| \cdot |\bar{\alpha}^{\bar{l}\zeta}| \\ &\leq |\bar{\alpha}^{\bar{l}\zeta}| \cdot 1.1 \cdot |\Lambda\zeta l_n| \leq |\bar{\alpha}^{\bar{l}\zeta}| \cdot e^{1.1\varepsilon_1 G} \cdot |\Lambda|, \end{aligned} \quad (17.4)$$

which contains a small factor Λ . Here we use the estimate

$$1.1|\zeta l_n| \leq R_{s,\nu} 2^{-s} L/A_n = \eta |\mathcal{X}_{s,\nu}|(T_s + 1)/A_n,$$

which together with (5.3), (11.5) implies that $1.1|\zeta l_n| < \exp(1.1\varepsilon_1 G)$. As a result, we get

$$\prod_{\sigma=1}^{\infty} |\Phi_s(z, \tilde{m}')|_{\sigma}^{1/\varkappa} = f(z) + \Delta_0,$$

where Δ_0 is the error produced by (9.4).

We apply the interpolation formula (16.1) to $f(z)$ and make the inverse substitution (9.4) in $f_1(z)$, $f_2(z)$. We get

$$\prod_{\sigma=1}^{\infty} |\Phi_s(z, \tilde{m}')|_{\sigma}^{1/\varkappa} = J_1 + \Delta_1 + J_2 + \Delta_2 + \Delta_0, \quad (17.5)$$

where J_1, J_2 are the corresponding expressions from (16.1), and Δ_1, Δ_2 are the errors produced by the inverse substitution (9.4).

The formula (16.1) for $f_2(z)$ contains the derivatives $f^{(\tau)}$. On account of (9.10) they satisfy

$$\begin{aligned} \frac{1}{\tau!} \Psi_s^{(\tau)}(x, \tilde{m}') &= \sum_{\|\tilde{\tau}'\|=\tau} \sum^{(s)} \sum_{l_0=0}^{L_0} p(\bar{l}') \tilde{\alpha}^{\tilde{\chi}(\bar{l}')x/b_n} \bar{\alpha}^{-\bar{v}_s x} \\ &\times \frac{1}{m_0! \tau_0!} 2^{-(S-s)m_0} \Delta^{(m_0+\tau_0)}(2^{S-s}z, l_0, H) \\ &\times d_0(m_0) \Delta(\tilde{\chi}(\bar{l}' - \bar{w}_s), \tilde{m}') (\tilde{\chi}(\bar{l}')/b_n)^{\tilde{\tau}} (\ln \tilde{\alpha})^{\tilde{\tau}} (\tilde{\tau}!)^{-\bar{1}}. \end{aligned} \quad (17.6)$$

Using this and Lemma 3.1, we see that the $\Psi_s^{(\tau)}(x, \tilde{m}')$ are linearly expressed in terms of $\Psi_s(x, \tilde{t}')$, $\|\tilde{t}'\| \leq \|\tilde{m}'\| + \tau$. The same holds for the inverse substitution (9.4), with Ψ replaced by Φ . Hence, by the inductive assumption (17.1), we have $J_2 = 0$ in (17.5).

The appearance of new terms after differentiation of the polynomial part was foreseen in Lemma 8.2. (In § 15 we use only the case $\tau = 0$, but in the choice of W_0 we assume that τ may increase.) The differentiation of the exponential part produces the expression in the last line of (17.6). This was foreseen in Lemma 8.3, and (15.4) shows that the parameter W_0 takes all new components into account. Therefore (8.5), (15.4) imply in (17.6) that

$$d_0(m_0) \sum_{\|\tilde{\tau}\| \leq M_0 - m_0} \left| \Delta(\tilde{\chi}(\bar{l}' - \bar{w}_s), \tilde{m}') (\tilde{\chi}(\bar{l}')/b_n)^{\tilde{\tau}} (\ln \tilde{\alpha})^{\tilde{\tau}} (\tilde{\tau}!)^{-\bar{1}} \right| \leq 0.01 e^G.$$

Since the bounds for x in Lemma 15.2 include all values of z, ζ needed in what follows, the estimate for the polynomial part is the same as in § 15.

There are some changes in the exponential part of (15.6). When we take the product over all σ , the component $|\bar{\alpha}|_{\sigma}^{\bar{v}_s z}$ disappears by the product formula. But we now consider not absolute heights (roots of degree D) but roots of degree $\varkappa$. Since the solid $\mathcal{W}$ is symmetric with respect to zero, all local heights of the exponential part are non-negative. Hence the absence of the factors with $\sigma = 1$, $\varkappa$ only simplifies the situation and enables us to use the above estimate of the exponential part:

$$\frac{1}{\varkappa} 2^{-s} L n |z| \leq \frac{1}{\varkappa} 2^{-s} L n 2^{\nu+1} X_s.$$

Moreover, in J_1 there appears an exponential factor containing ζ . By condition (10.3) this factor is estimated as $\exp(\eta^{-1} 2^{-s} L |\zeta|)$.

To estimate the interpolation part $|Q(z)/Q(\zeta)|$ in J_1 we use (16.2), and $\deg Q = |\mathcal{X}_{s,\nu}|(T_s + 1)$ is estimated by the two-sided inequality (11.5).

We note that if $z \in \mathcal{X}_{s,\nu+1}$, then we have $z_0 \leq |\mathcal{X}_{s,\nu}|$ in Lemma 16.1. Combining the estimates in Lemmas 15.1, 15.2 for the components of J_1 , taking (13.2), (11.5)

into account and putting $|\zeta| = R = R_{s,\nu}$, we get

$$\begin{aligned}
 \ln(8|J_1|) &\leq \frac{1}{\varkappa}(2^{\nu+1}X_s)(2^{-s}L)n + \frac{D}{\varkappa}(2 + n\varepsilon_1)G \\
 &\quad + \eta^{-1}2^{-s}LR_{s,\nu} + |\mathcal{X}_{s,\nu}|(T_s + 1)\ln\left(\frac{|\mathcal{X}_{s,\nu}|}{R_{s,\nu}}\right) \\
 &\leq \frac{1}{\varkappa}|\mathcal{X}_{s,\nu}|(T_s + 1)n + \frac{D}{\varkappa}2.1G \\
 &\quad - |\mathcal{X}_{s,\nu}|(T_s + 1)\ln\left(e^{(n+0.53)/\varkappa}\frac{(T_s + 1)}{2^{-s}L}\right) \\
 &\leq \frac{1}{\varkappa}|\mathcal{X}_{s,\nu}|(T_s + 1)\left(n + \frac{1}{4} \cdot 2.1 - (n + 0.53)\right) < 0, \quad (17.7)
 \end{aligned}$$

that is, $|J_1| < 1/8$. We see from (17.4), (12.1) that $|\Delta_1| \leq 2|J_1| < 1/4$. These inequalities hold mainly due to the choice of the parameter η in (10.3).

The smallness of Δ_0, Δ_2 ($< 1/4$) will follow from the assumption (12.2) (that $|\Lambda|$ is small) and will be verified in the next section. Here $J_2 = 0$. Then (17.5) yields the inequality (17.2), which proves the inductive step.

§ 18. Division of the argument

In this section we prove the inductive step in s , that is, we assume equations (11.6) (equivalently, (17.1)) for (s, n) and get the same equations for $(s + 1, 0)$. We first obtain the equations

$$F_s\left(\frac{x}{2}, \tilde{m}'\right) = 0, \quad \|\tilde{m}'\| \leq M_{s+1}, \quad x \in \mathcal{X}_{s+1,0} \quad (18.1)$$

and then deduce (17.1) using Kummer theory.

We prove (18.1) at the end of the section. Now we present some corollaries of (18.1). Consider the construction of the sets $\mathcal{L}_s$. The numbers $\alpha_1, \dots, \alpha_n$ correspond to vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ (linearly independent over $\mathbb{Z}$ by the hypothesis of Theorem 2.1) in the extended logarithmic space. They generate a lattice M_n with basis $\mathbf{e}_1, \dots, \mathbf{e}_n$. These vectors correspond to numbers $\vartheta_1, \dots, \vartheta_n$ in the field $\mathbb{K}$, which have an important algebraic property.

Lemma 18.1. *Suppose that $\mathbb{K}$ contains a p th root of unity (p being a prime). Then $\vartheta_1, \dots, \vartheta_n$ satisfy*

$$[\mathbb{K}(\vartheta_1^{1/p}, \dots, \vartheta_n^{1/p}) : \mathbb{K}] = p^n.$$

Proof. This lemma is essentially proved in [15] (see also [4]). But since it is not explicitly stated there, we give a proof.

Since $\mathbb{K}$ contains a p th root of unity, we can choose the value of $\vartheta_i^{1/p}$ arbitrarily. We assume this choice to be consistent with that of $\ln \vartheta_i$. The proof is by induction on j , where we consider the fields

$$\mathbb{K}_0 = \mathbb{K} = \mathbb{Q}(\vartheta_1, \dots, \vartheta_n), \quad \mathbb{K}_j = \mathbb{K}(\vartheta_1^{1/p}, \dots, \vartheta_j^{1/p}), \quad 1 \leq j \leq n.$$

We must prove that $[\mathbb{K}_{j+1} : \mathbb{K}_j] = p$. If this does not hold for some j , then there are $\alpha \in \mathbb{K}$ and $r_i \in \{0, 1, \dots, p-1\}$, $1 \leq i \leq j$ such that

$$\vartheta_{j+1} = \vartheta_1^{r_1} \dots \vartheta_j^{r_j} \alpha^p$$

(see [15], Lemma 3). Then the vector in the extended logarithmic space corresponding to the number α with the natural choice of the logarithm is linearly dependent on $\mathbf{e}_1, \dots, \mathbf{e}_n$ over $\mathbb{Z}$ but is not integrally expressible in terms of them, thus contradicting the hypothesis that the $\mathbf{e}_j$ form a basis of M_n . This proves the lemma.

We use Lemma 18.1 for $p = 2$, which is consistent with (1.2) and preempts the need for a root of unity (here equal to -1).

We return to the proof of the theorem. Coordinates in M_n correspond to multi-powers of the corresponding sets of numbers, and by (10.1) we have

$$\bar{\alpha}^{\bar{\lambda}} = \bar{\vartheta}^{\bar{\mu}}, \quad \bar{\lambda} \in \mathcal{M}_n, \quad \bar{\mu} \in \mathbb{Z}^n, \quad \bar{\lambda} = \bar{\mu} \mathcal{A}^{-1}, \quad (18.2)$$

where $\mathcal{A}$ is the transition matrix from the basis $\{\mathbf{a}_i\}$ to the basis $\{\mathbf{e}_j\}$. We now put

$$x = 2y + 1, \quad \bar{\mu} = 2\bar{\tau} + \bar{\delta}, \quad \bar{\delta} \in \{0, 1\}^n, \quad \bar{\tau} \in \mathbb{Z}^n.$$

Taking (18.2) into account and omitting the common factor $\bar{\alpha}^{\bar{\nu}x/2}$, we see from (18.1) that

$$\begin{aligned} 0 &= F_s\left(\frac{x}{2}, \tilde{m}'\right) = \sum_{\bar{\mu}}^{(s)} P_s^{(m_0)}\left(\frac{x}{2}, \bar{l}\right) \tilde{\chi}^{\tilde{m}}(\bar{l}) \bar{\vartheta}^{\bar{\mu}x/2} \\ &= \sum_{\bar{\delta}} \left(\sum_{\bar{\tau}} P_s^{(m_0)}\left(\frac{x}{2}, \bar{l}\right) \tilde{\chi}^{\tilde{m}}(\bar{l}) \bar{\vartheta}^{\bar{\tau}x + \bar{\delta}y} \right) \bar{\vartheta}^{\bar{\delta}/2}. \end{aligned} \quad (18.3)$$

At this point we use the Kummer condition that the numbers $\bar{\vartheta}^{\bar{\delta}/2}$, $\bar{\delta} \in \{0, 1\}^n$, are linearly independent over $\mathbb{K}$. Since the coefficients of $\bar{\vartheta}^{\bar{\delta}/2}$ in (18.3) belong to $\mathbb{K}$, they all vanish, and we see that the terms in (18.3) are equal to zero for each value of $\bar{\delta}$. Passing back to the vector $\bar{\alpha}$ with the help of (18.2), we get

$$\begin{aligned} 0 &= \sum_{\bar{\tau}}^{(s+1)} P_s^{(m_0)}\left(\frac{x}{2}, \bar{l}\right) \tilde{\chi}^{\tilde{m}}(\bar{l}) \bar{\vartheta}^{\bar{\mu}x/2} \\ &= \sum_{\bar{\tau}}^{(s+1)} P_s^{(m_0)}\left(\frac{x}{2}, \bar{l}\right) \tilde{\chi}^{\tilde{m}}(\bar{l}) \bar{\alpha}^{\bar{\lambda}x/2}. \end{aligned} \quad (18.4)$$

We shall work only with the $\bar{\delta}$ corresponding to the initially chosen value of $\bar{l}$ such that there is a non-zero $p(l_0, \bar{l})$. Redefining $\bar{\lambda}/2$ and taking into account that $\bar{\tau}$ is expressed in terms of $\bar{\lambda}$, we may assume that the sum in (18.4) runs over $\bar{\lambda} \in \mathcal{L}_{s+1}$. This is consistent with the requirements on $\mathcal{L}_{s+1}$ in §§ 9, 10 since we obviously have $\mathcal{L}_{s+1} \subseteq \mathcal{L}_s/2$ and

$$\frac{1}{2}\bar{\lambda} = \frac{1}{2}\bar{\mu}\mathcal{A}^{-1} = \bar{\tau}\mathcal{A}^{-1} + \frac{1}{2}\bar{\delta}\mathcal{A}^{-1} \in \mathcal{M}_n + \frac{1}{2}\bar{\delta}\mathcal{A}^{-1}.$$

We leave the indices $\bar{l}$ unchanged in the polynomial and differential parts, having in mind that they are expressed in terms of $\bar{\lambda}$ and $\bar{\delta}$. Multiplying the expression in (18.4) again by $\bar{\alpha}^{\bar{\nu}x/2}$ and denoting it by F_{s+1} , we see that condition (11.6) holds for $s+1$ and $\nu=0$.

We now turn to the proof of (18.1), where the functions F may be replaced by Φ . In contrast to §17, we must work in the field $\mathbb{K}_n$ of degree $2^n D$. Label its valuations by indices σ and regard $f(\zeta)$ in (17.3) as a function defined over $\mathbb{K}_n$. Here the parameter z is

$$z = \frac{1}{2}x, \quad x \in \mathcal{X}_{s+1}, \quad |z| \leq X_{s+1}.$$

In (17.5) we similarly define $\Delta_0, J_1, \Delta_1, J_2, \Delta_2$ with $\nu = n$. Here again $J_2 = 0$. In Lemma 16.1 we put $T = T_s$, $\mathcal{X} = \mathcal{X}_{s,n}$, $\delta = 1$ (the common difference), $r = 1/4$, and $z_0 = |\mathcal{X}_{s,n}|/2$, the radius $R = R_{s,n}$ being defined in §12. The estimate for J_1 is obtained as in (17.7) if we take into account the new bound for z ($|z| = |x|/2 \leq X_{s+1}$, see (11.4)) and the degree of the field:

$$\begin{aligned} \ln(8|J_1|) &\leq \frac{2^n}{\varkappa} 2^{-s} L n X_{s+1} + \frac{2^n D}{\varkappa} (2 + n\varepsilon_1) G \\ &\quad + \eta^{-1} 2^{-s} L R_{s,n} + |\mathcal{X}_{s,n}| (T_s + 1) \ln(|\mathcal{X}_{s,n}|/2R_{s,n}) \\ &\leq \frac{2^n}{\varkappa} 2^{-s} L n X_{s+1} + \frac{2^n D}{\varkappa} (2 + n\varepsilon_1) G \\ &\quad - |\mathcal{X}_{s,n}| (T_s + 1) \ln \left(e^{(n+0.53)/\varkappa} \frac{(T_s + 1)}{2^{1-s} L} \right) \\ &\leq \frac{D}{\varkappa} 2^{n+2} G \left(n(1 + \varepsilon_2) + \frac{1}{4} (2 + n\varepsilon_1) - (n + 0.53) \right) < 0, \quad (18.5) \end{aligned}$$

whence $|J_1| < 1/8$ and $|\Delta_1| < 1/4$ (see §17). This is again mainly due to the choice of η .

It remains to prove that the assumption (12.2) guarantees that $|\Delta_0|, |\Delta_2| < 1/4$. Then (17.2) holds, which proves the inductive step. At all inductive steps we take the worst case occurring as we pass from s to $s+1$.

We estimate Δ_2 similarly to J_1 but with the following differences. For J_1 we majorize the number $|f(\zeta)|$, $|\zeta| \leq R_{s,\nu}$, in the formula for $f_1(z)$ in (16.1). For Δ_2 we majorize $|f^{(\tau)}(x)|$, $|x| \leq 2^n X_s$. It follows from (17.6) that the estimate for $|f^{(\tau)}(x)|/\tau!$ is the same as for $|f(\zeta)|$ (that is, (17.7) for $\nu < n$ and (18.5) for $\nu = n$) but with $R_{s,\nu}$ replaced by $2^n X_s$. To estimate the interpolation part, we replace (16.2) by the inequality (16.3), which takes the following form in view of the formula for z_0 :

$$\begin{aligned} \ln |Q(z)/Q(\zeta)| &\leq |\mathcal{X}_{s,\nu}| (T_s + 1) \ln(2e), & \nu < n, \\ \ln |Q(z)/Q(\zeta)| &\leq |\mathcal{X}_{s,\nu}| (T_s + 1) \ln(e), & \nu = n. \end{aligned}$$

Thus the second term (corresponding to (16.2)) in the second line of the inequalities (17.7), (18.5) must be replaced by the estimate above.

Moreover, we add the factor from (17.4) to the estimate:

$$|\Lambda| R_{s,n} 2^{-s} L / A_n < |\Lambda| e^{1.1\varepsilon_1 G}.$$

And finally we must estimate the sum in f_2 in (16.1). This was not needed before because of the condition $J_2 = 0$. Since $r \leq 1/2$ in all cases, the sum over τ in (16.1) is estimated by 2 on account of the term r^τ in the integral. Hence the sum in f_2 gives $4|\mathcal{X}| \leq e^{1.1\varepsilon_1 G}$. Taking the worst case, $\nu = n$, we get

$$\begin{aligned} \ln(8|\Delta_2|) &\leq \frac{1}{\varkappa} 2^{-s} L n 2^n X_{s+1} + \frac{D}{\varkappa} 2^n (2 + 3n\varepsilon_1) G \\ &\quad + \eta^{-1} 2^{-s} L 2^n X_s + |\mathcal{X}_{s,n}| (T_s + 1) + \ln |\Lambda| \\ &\leq \frac{D}{\varkappa} 2^{n+1} G (1 + 2\varepsilon_2) \left(2n + 1 + 2n\varepsilon_1 + \frac{\varkappa}{\eta} + 2\varkappa \right) + \ln |\Lambda|. \end{aligned}$$

On account of (11.5), (12.1) and the assumption (12.2) it follows that $\ln |8\Delta_2| < 0$. For Δ_0 we similarly get $|\Delta_0| < 1/2$. Using (17.5), (17.2), we complete the reasoning on extrapolation of zeros and prove the inductive step.

§ 19. The zero estimate

In this section we prove the zero estimate for exponential quasipolynomials in several variables. This is an adapted version of a result of Wüstholz [19] for the case of homogeneous linear forms in logarithms of algebraic numbers. The statement takes into account some delicate points mentioned in [4], [17], [20]. We revise the geometric part of the argument of [19], [4], which enables us to improve the constant $\mathcal{C}(\nu)$ (see Proposition 19.2). In the statement we use a symbolic notation, whose relation to the parameters of the preceding sections is established in § 20. Moreover, to state and prove assertions in this section, we need some notions concerning algebraic groups (see [19], [20]).

Algebraic subgroups in $\mathbb{G}_a \times \mathbb{G}_m^n$ are of the form $\mathbb{H} = \mathbb{H}_a \times \mathbb{H}_m$, $\mathbb{H}_a \subseteq \mathbb{G}_a$, $\mathbb{H}_m \subseteq \mathbb{G}_m^n$, where either $\mathbb{H}_a = \{0\}$ or $\mathbb{H}_a = \mathbb{G}_a$. Furthermore, if $\nu > 0$, then the connected group $\mathbb{H}_m$ of rank ν is defined by relations of the form $\overline{Y}^{\vec{l}} = 1$, where $\vec{l} \in M$ and $M = \langle \vec{v}_1, \dots, \vec{v}_\nu \rangle_{\mathbb{Z}} \subseteq \mathbb{Z}^n$ is a lattice of rank ν . Then $r = \text{rank } \mathbb{H} = \nu + \varepsilon$, where ε equals 0 or 1 depending on $\mathbb{H}_a$. We put

$$d(j_1, \dots, j_\nu) = |\det(v_{i,j_k})_{1 \leq i, k \leq \nu}|, \quad 1 \leq j_1 < \dots < j_\nu \leq n.$$

These expressions are independent of the choice of basis in M . We also need the following estimate for the coefficients $\delta(\mathbb{H}; i_1, \dots, i_r)$ of the Hilbert polynomial associated with $\mathbb{H}$:

$$\begin{aligned} \delta(\mathbb{H}; 0, i_2, \dots, i_r) &\geq d(i_2, \dots, i_r), \quad r = \nu + 1, \\ \delta(\mathbb{H}; i_1, \dots, i_r) &\geq d(i_1, \dots, i_r), \quad r = \nu. \end{aligned}$$

Suppose that $\vec{b} = (b_1, \dots, b_n)$ is a set of numbers, $b_n \neq 0$, and $\vec{\vartheta} = (\vartheta_1, \dots, \vartheta_n)$ is an n -tuple of non-zero algebraic numbers. In the ring of polynomials $\mathbb{C}[Y_0, \dots, Y_n]$

we consider differential operators ∂_j as defined in (3.1). Suppose that there are integers

$$\begin{aligned} L'_0, \dots, L'_n &\geq 0 \quad (\text{degrees}), \\ X'_0 &\geq \dots \geq X'_n \geq 0, \quad X'_0 + \dots + X'_n \leq X' \quad (\text{zeros}), \\ M'_0 &\geq \dots \geq M'_n \geq 0, \quad M'_0 + \dots + M'_n \leq M' \quad (\text{multiplicities}) \end{aligned}$$

with the additional condition that the following inequality holds for all $0 \leq \nu \leq n$ and all $\overline{m}' = (m_0, \dots, m_n) \in \{0, 1\}^{n+1}$, $\|\overline{m}'\| = \nu + 1$ ($\|\cdot\|$ is the sum-norm):

$$(X'_\nu + 1) \binom{M'_\nu + \nu + 1 - \delta_{\nu,n}}{\nu + 1 - \delta_{\nu,n}} > (\nu + 1)! L_0'^{m_0} \dots L_n'^{m_n}, \quad (19.1)$$

where $\delta_{\nu,n}$ is the Kronecker symbol and $0^0 = 1$.

We distinguish the following cases for the group $\mathbb{H}$.

- (i) $\text{Lie } \mathbb{H} \subseteq (\partial_0, \dots, \partial_{n-1})$. Then we put $\sigma = r - 1$; otherwise $\sigma = r$.
- (ii) There is $x \in \{1, \dots, X'\}$ such that $(x, \vartheta_1^x, \dots, \vartheta_n^x) \in \mathbb{H}(\overline{\mathbb{Q}})$. Then we put $E = 1$; otherwise $E = X'_{r-1} + 1$.

Proposition 19.1 ([19], Theorem 1.1). *In this notation, suppose that there is a polynomial $P \in \mathbb{C}[Y_0, \dots, Y_n]$ with $P \neq 0$, $\deg_{Y_j} P \leq L'_j$, $0 \leq j \leq n$, such that*

$$\partial_0^{\mu_0} \dots \partial_{n-1}^{\mu_{n-1}} P(x, \vartheta_1^x, \dots, \vartheta_n^x) = 0 \quad (19.2)$$

for $x = 0, \dots, X'$ and for all $\tilde{\mu}' \in \mathbb{N}_0^n$ with $\|\tilde{\mu}'\| \leq M'$.

Then there is a connected algebraic group $\mathbb{H} \subseteq \mathbb{G}_a \times \mathbb{G}_m^n$ of rank $r \leq n$ such that for all $0 \leq i_1 < \dots < i_r \leq n$ we have

$$\begin{aligned} \delta(\mathbb{H}; i_1, \dots, i_r) &\leq \gamma_r L'_{i_1} \dots L'_{i_r}, \\ \gamma_r &= r! E^{-1} \binom{M'_{r-1} + \sigma}{\sigma}^{-1}. \end{aligned} \quad (19.3)$$

We now state the zero estimate which is used to complete the proof of the main theorem. We begin by introducing some additional restrictions that naturally appear in the previous sections. They constrain the possible cases of Proposition 19.1.

We have a set $\bar{A} = (A_1, \dots, A_n)$ of positive numbers and fix the values of $\ln \vartheta_1, \dots, \ln \vartheta_n$ such that the logarithms are linearly independent over $\mathbb{Z}$ and the following inequalities hold:

$$|b_1 \ln \vartheta_1 + \dots + b_n \ln \vartheta_n| < \frac{2\pi}{X' + 1}, \quad (19.4)$$

$$M'_0 + 1 > L'_j, \quad 1 \leq j \leq n, \quad (19.5)$$

$$n(n+1)L'_0 > M'_n + n. \quad (19.6)$$

Proposition 19.2. *If conditions (19.1), (19.2), (19.4)–(19.6) hold, then one can find ν ($1 \leq \nu < n$) and a set of independent vectors $\overline{Z}_i \in \mathbb{Z}^n$, $1 \leq i \leq \nu$, such that $\bar{b} \in \langle \overline{Z}_1, \dots, \overline{Z}_\nu \rangle_{\mathbb{Q}}$ and the numbers*

$$A'_i = \|\overline{Z}_i\|_A = A_1|Z_{i,1}| + \dots + A_n|Z_{i,n}|, \quad 1 \leq i \leq \nu,$$

satisfy at least one of the inequalities

$$(X'_{\nu-1} + 1) \binom{M'_{\nu-1} + \nu - 1}{\nu - 1} A'_1 \dots A'_\nu \leq \mathcal{C}(\nu) \max\{\overline{A}^{\overline{m}} \overline{L}'^{\overline{m}}\}, \quad (19.7)$$

$$(X'_\nu + 1) \binom{M'_\nu + \nu}{\nu} A'_1 \dots A'_\nu \leq \mathcal{C}(\nu)(\nu + 1)L'_0 \max\{\overline{A}^{\overline{m}} \overline{L}'^{\overline{m}}\}, \quad (19.8)$$

where the maximum is taken over all $\overline{m} \in \{0, 1\}^n$ with $\|\overline{m}\| = \nu$, and

$$\mathcal{C}(\nu) = 2^\nu \left(\frac{2n}{\pi} \right)^{\nu/2} \left(\frac{\nu}{2} \right)! \nu! \left(\frac{n}{\nu} \right)^{1/2}. \quad (19.9)$$

Remark. In [4] one has $\mathcal{C}(\nu) = 2^{1-\nu}(\nu!)^2(n\nu)^\nu$ and one requires that the set of vectors $\overline{Z}_i$ be primitive.

Proof. As in [4], we first exclude some possibilities in Proposition 19.1. In the case (ii) we have

$$l_1 \ln \vartheta_1 + \dots + l_n \ln \vartheta_n = \frac{2\pi i}{x} l_0, \quad 1 \leq x \leq X', \quad l_0 \in \mathbb{Z}, \quad \bar{l} \in M,$$

where M is the lattice (of rank $\nu = r$) defining the group $\mathbb{H}_m$. Since the $\ln \vartheta_j$ are linearly independent, we have $\nu = 1$ (otherwise $2\pi i$ may be excluded) and $l_0 \neq 0$. We may assume that the vector $\bar{l}$ is primitive. If we also have case (i), then $\bar{b} = m_0 \bar{l}$, $m_0 \in \mathbb{Z}$. But this contradicts condition (19.4). Hence (i) does not hold, and we have $\sigma = r = \nu = 1$, $E = 1$, $\gamma_r = (M'_0 + 1)^{-1}$. But then (19.3) contradicts condition (19.5) since we have $\delta(\mathbb{H}; i) \geq 1$ for some $1 \leq i \leq n$. Thus the case (ii) is completely excluded, and $E = X'_{r-1} + 1$.

If the case (i) does not hold, then $\sigma = r$ and a consideration of $\delta(\mathbb{H}; i_1, \dots, i_r) \geq 1$ shows that (19.3) contradicts (19.1). In the case (i) we have $\bar{b} \in \langle M \rangle_{\mathbb{Q}}$, whence $\nu \geq 1$. Assume that $\nu = n$. Since $\nu \leq r \leq n$, we have $r = n = \sigma$ and a consideration of $\delta(\mathbb{H}; 1, \dots, n) \geq 1$ shows that (19.3) contradicts (19.1) for $\nu = n$ (on account of (19.6) and the monotonicity of X'_j , M'_j).

Thus we obtain the required lattice $M \subseteq \mathbb{Z}^n$ of rank ν ($1 \leq \nu < n$). It remains to prove that there are independent vectors $\overline{Z}_1, \dots, \overline{Z}_\nu \in M$ satisfying (19.7) or (19.8).

The space $\mathcal{V}_n = \mathbb{R}^n$ is endowed with the sum-norm $\|\cdot\|_A$. Its unit ball is the octahedron $\mathcal{W}_n$, given by $|x_1|A_1 + \dots + |x_n|A_n \leq 1$ in the standard coordinates $(x_1, \dots, x_n)$. We introduce new coordinates $x'_j = A_j x_j$, $1 \leq j \leq n$ in $\mathcal{V}_n$ and assume that they are Euclidean. Then the matrix of the corresponding quadratic form in the old coordinates $\{x_j\}$ is $Q = \text{diag}\{A_1^2, \dots, A_n^2\}$. We denote this metric by $|\cdot|_Q$:

$$|\mathbf{v}|_Q^2 = x_1'^2 + \dots + x_n'^2 = (x_1, \dots, x_n)Q(x_1, \dots, x_n)^T.$$

The octahedron $\mathcal{W}_n$ is given by $|x'_1| + \dots + |x'_n| \leq 1$ in the new coordinates. Thus it is regular. We easily see that the Q -Euclidean ball $\mathcal{S}$, defined by

$$|\mathbf{v}|_Q^2 = x_1'^2 + \dots + x_n'^2 \leq R^2,$$

is contained in $\mathcal{W}_n$ if $R = 1/\sqrt{n}$.

To estimate the ν -dimensional Q -Euclidean volume of the lattice M , we choose a basis $\bar{v}_1, \dots, \bar{v}_\nu$ of M . Let V be the matrix with rows formed by the coordinates of the basis vectors. Applying the Binet–Cauchy formula and (19.3) with $\sigma = r - 1$ (case (i)), we get

$$\begin{aligned} \text{Vol}_\nu^2 M &= \det(VQV^T) = \sum A_{j_1}^2 \dots A_{j_\nu}^2 d^2(j_1, \dots, j_\nu) \\ &\leq \sum (A_{j_1} \dots A_{j_\nu} \gamma_r L_0'^\varepsilon L_{j_1}' \dots L_{j_\nu}')^2 \leq \binom{n}{\nu} (\gamma_r L_0'^\varepsilon \max\{\bar{A}^{\bar{m}} \bar{L}'^{\bar{m}}\})^2. \end{aligned}$$

Here the sum is taken over the sets $1 \leq j_1 < \dots < j_\nu \leq n$, the maximum is taken over $\bar{m} \in \{0, 1\}^n$, $\|\bar{m}\| = \nu$, and ε depends on $\mathbb{H}_a$.

We now apply Proposition 4.1 to the lattice M , the norm $\|\cdot\|_A$ and the Euclidean metric $|\cdot|_Q$. We work in the ν -dimensional space $\mathcal{V}_\nu = \langle M \rangle_{\mathbb{R}}$. Here the unit ball is $\mathcal{W}_\nu(\|\cdot\|_A) = \mathcal{W}_n \cap \mathcal{V}_\nu$. Estimate its Q -Euclidean volume by the inequality

$$\text{Vol}_\nu \mathcal{W}_\nu(\|\cdot\|_A) \geq \text{Vol}_\nu(\mathcal{S}_R \cap \mathcal{V}_\nu) = R^\nu (\pi/2)^{\nu/2} / (\nu/2)!.$$

Then Proposition 4.1 yields the existence of independent vectors $\bar{Z}_1, \dots, \bar{Z}_\nu \in M$ such that

$$\begin{aligned} \|\bar{Z}_1\|_A \dots \|\bar{Z}_\nu\|_A &\leq 2^\nu \frac{\text{Vol}_\nu M}{\text{Vol}_\nu \mathcal{W}_\nu(\|\cdot\|_A)} \\ &\leq 2^\nu \left(\frac{2n}{\pi}\right)^{\nu/2} \left(\frac{\nu}{2}\right)! \binom{n}{\nu}^{1/2} \gamma_r L_0'^\varepsilon \max\{\bar{A}^{\bar{m}} \bar{L}'^{\bar{m}}\}. \end{aligned}$$

This yields one of the inequalities (19.7), (19.8), depending on the group $\mathbb{H}$: if $r = \nu$ we get (19.7), and if $r = \nu + 1$ we get (19.8). This proves Proposition 19.2. Note that we did not use Mahler's lemma 4.2.

§ 20. Proof of the main theorem

Provided that the linear form in logarithms is small (see (12.2)), we have constructed a non-zero solution of the system (9.9) for $(s, \nu) = (S, n)$. In what follows ν is not used in this meaning. The system (9.9) is equivalent to (9.5), that is, (9.9) is of the form (19.2) with the polynomial $P = \mathcal{F}_S$ from (9.1).

As mentioned in the introduction, we improve the estimates of linear forms in two aspects concerning the parameters A_j . Firstly, there are no lower bounds for A_j . Thus A_j may be small. (This improvement of [4] was partially done in [1].) Secondly, we remove the factors A_j from the parameter C_0 . (This was done in [4] by Proposition 19.1 but was not done in [1].) These results are of different kinds, since the presence of $A_j < 1$ in C_0 only improves the estimate of $|\Lambda|$.

Hence it would be better to retain such factors in C_0 , but we remove them to simplify and unify the exposition.

Since the possible smallness of Ω still creates some difficulties in the application of Proposition 19.2, we begin by considering the simpler case when

$$C_0\omega < e^{-3}n^2c_1/3.$$

Then the definition of S and estimate (5.3) imply that the sum over l_j , $1 \leq j \leq n$, disappears in $\mathcal{F}_S$ since the range for these indices becomes $< 1/N$. Thus $\mathcal{F}_S$ simply becomes a polynomial in Y_0 . We see from (11.5) that the number of zeros of $\mathcal{F}_S$ counting multiplicities is at least

$$|\mathcal{X}_{S,n}|(T_S + 1) \geq 2^{n+2}G > L_0,$$

which contradicts the condition $\mathcal{F}_S \neq 0$ and hence proves the theorem in the case when ω is small. In what follows we thus assume that

$$C_0\omega \geq e^{-3}n^2c_1/3. \quad (20.1)$$

If (20.1) holds, the lower bound for X_S can be increased:

$$X_S \geq 0.5e^{-3}n^2c_1\xi DW_0.$$

If $T_S = 0$, this inequality follows directly from (11.3) and (20.1). If $T_S \geq 1$, it is obtained from the definition of S since

$$T_S + 1 \leq 2^{1-S}C_0\xi\omega/c_1.$$

As noted in § 9, a linear change of variables makes all exponents integral and non-negative and shifts the zeros to the non-negative domain, where they take the form $x = 0, \dots, X'$.

We now apply Proposition 19.2. To do this, we relate the parameters of § 19 to the previous variables. Since the exponents in the function $\mathcal{F}$ are fractional with denominator N , we must define the numbers ϑ_j by the equations $\ln \vartheta_j = \frac{1}{N} \ln \alpha_j$. (The numbers remain algebraic, and the degree of the field is irrelevant in Proposition 19.2.) We now put

$$\begin{aligned} X' &= |\mathcal{X}_{S,n}| - 1 \geq 2^{n+1} [0.5e^{-3}n^2\xi c_1 DW_0], \\ X'_\nu &= \left\lfloor \frac{X'}{n+1} \right\rfloor, \quad 0 \leq \nu \leq n, \\ M' &= M_{S,n} \geq \frac{\varepsilon_0 \xi C_0 \omega}{1 + \varepsilon_0} = \frac{6n^3 C_0 \omega (n+1)}{c_1}, \\ M'_\nu &= \left\lfloor \frac{M'}{n+1} \right\rfloor, \quad 0 \leq \nu \leq n, \\ L'_0 &= L_0, \quad L'_j = \left\lfloor 2^{-S} \frac{NL}{A_j} \right\rfloor, \quad 1 \leq j \leq n, \end{aligned}$$

and estimate these parameters. Taking the orders of c_1 and ε_0 into account, we have

$$X'_\nu \geq \frac{X'}{n+1} - 1 \geq \frac{1}{n+1} 2^n e^{-3} n^2 \xi_{c_1} DW_0 - 1 \geq 2^{n+1} e^{-3} n^2 \xi DW_0,$$

$$M'_\nu + 1 \geq \frac{6n^3 C_0 \omega}{c_1}, \quad 0 \leq \nu \leq n.$$

The formula for S and inequality (5.3) imply that

$$L'_j \leq 2^{-S} \frac{NL}{A_j} \leq \frac{3e^3}{n^2 c_1} C_0 \omega \min \left\{ 1, \frac{1}{1.5 A_j} \right\}, \quad 1 \leq j \leq n, \quad (20.2)$$

whence, in particular, $L'_j \leq M'_j$.

We now verify that the other hypotheses of Proposition 19.1 hold.

Since $W_0 > C_0/2$ and ε_0 is sufficiently small, we have $L'_0 > M' + 1$ and $L'_0 \geq L'_j$, $j \geq 1$. To simplify the verification of (19.1), we order the degrees L'_j as we please: for example, assume that

$$L'_0 \geq L'_1 \geq \dots \geq L'_n.$$

This enables us to take $m_0 = \dots = m_\nu = 1$, $m_{\nu+1} = \dots = m_n = 0$ in (19.1). We may also assume that $m_j = 0$ whenever $L'_j = 0$, otherwise (19.1) is trivial. We now begin to verify that (19.1) holds. If $\nu < n$, then it suffices to verify that

$$(X'_\nu + 1) \left(\frac{M'}{n+1} \right)^{\nu+1} > ((\nu+1)!)^2 L'_0 L'_1 \dots L'_\nu.$$

If we substitute the estimates of X'_ν , M' , L_0 , L'_j and replace $(\nu+1)!$ by $e^{-\nu}(\nu+1)^{\nu+1.5}$, we get

$$\frac{2^{n+1} n^2 \xi DW_0}{e^3} \left(\frac{6n^3 C_0 \omega}{c_1} \right)^{\nu+1} \geq e^{-2\nu} (\nu+1)^{3+2\nu} W_0 \xi \omega \left(\frac{3e^3 C_0 \omega}{n^2 c_1} \right)^\nu.$$

After cancelling equal factors and taking logarithms, it remains to verify that

$$\ln \left(\frac{3 \cdot 2^{n+2} n^5 DC_0}{e^3 (\nu+1)^3 c_1} \right) + \nu \ln \left(\frac{6n^5}{3e(\nu+1)^2} \right) \geq 0.$$

The left-hand side is convex upwards with respect to ν . Hence it suffices to verify the inequality for $\nu = 0$ and $\nu = n-1$, which is elementary.

For $\nu = n$, on account of the factors A_j in (20.2), we must verify that

$$\frac{2^{n+1} n^2 \xi DW_0}{e^3} \left(\frac{6n^3 C_0 \omega}{c_1} \right)^\nu \geq e^{2-2n} (n+1)^2 n^{2n} W_0 \xi \omega \left(\frac{3e^3 C_0 \omega}{n^2 c_1} \right)^n \frac{1}{1.5^n \Omega}.$$

Substituting the formula for ω and cancelling equal terms, we get

$$\frac{2n^2}{e^5 (n+1)^2} \left(\frac{6n^3}{e} \right)^n \geq \frac{\gamma}{N}.$$

This inequality is valid because $\gamma \leq N$ and $n \geq 2$.

To verify hypothesis (19.4), note that

$$X' \leq 2^{n+1} e^{-3} n^3 \xi_{c_1} DW_0 \leq e^{C_0} e^{W_0/e} \leq e^{2\varepsilon_1 G},$$

and the assumption (12.2) implies that

$$|b_1 \ln \vartheta_1 + \dots + b_n \ln \vartheta_n| = \frac{1}{N} |\Lambda| < e^{-4\varepsilon_1 G} < \frac{2\pi}{X' + 1}.$$

The conditions (19.5), (19.6) are also easily verified.

Thus all conditions of § 19 hold and Proposition 19.2 may be applied. If we show that (7.3) follows from each of the inequalities (19.7), (19.8), then we get a contradiction to the inductive assumption of § 7 and thus prove Theorem 2.1.

The numbers $A_j L'_j$, $j \geq 1$, are estimated in the same way and do not exceed L'_0 . Hence we may assume in (19.7), (19.8) that

$$m_1 = \dots = m_\nu = 1, \quad m_{\nu+1} = \dots = m_n = 0.$$

Since all X'_ν are equal and equal to M'_ν and since we have

$$W_0(M'_\nu + \nu)/\nu \leq (\nu + 1)C_0 L'_0,$$

it suffices to consider only (19.8) from the equations (19.7), (19.8), or

$$\frac{2^{n+1} n^2 \xi DW_0}{e^3} \left(\frac{6n^3 C_0 \omega}{c_1} \right)^\nu A'_1 \dots A'_\nu \geq (\nu + 1)! \mathcal{C}(\nu) W_0 \xi \omega \left(\frac{2e^3 C_0 \omega}{n^2 c_1} \right)^\nu.$$

Cancelling equal terms, substituting (19.9) for $\mathcal{C}(\nu)$ and using the estimates $\binom{n}{\nu} \leq n^\nu/\nu!$ and $x! \leq e\nu^{\nu+1/2}e^{-\nu}$, we get

$$A'_1 \dots A'_\nu \leq \frac{\gamma}{N} g(\nu) \Omega, \quad g(\nu) = \frac{e^{11/2}(\nu+1)\nu^{5/4}}{2^{n+3/2}n^2} \left(\frac{4e^2\nu^4}{9\pi n^8} \right)^{\nu/2}. \quad (20.3)$$

We easily verify that

$$g(\nu) \leq \frac{1}{2} (4e^2)^{n-\nu}, \quad 1 \leq \nu \leq n.$$

But then (20.3) yields condition (7.3). This contradicts the inductive assumption and proves the theorem.

We mention another novel feature in the proof of Theorem 2.1. The numbers $\vartheta_1, \dots, \vartheta_n$ in Lemma 18.1 were used only to justify division of the argument. Other calculations used $\alpha_1, \dots, \alpha_n$ directly. All previous papers contained a passage to ϑ_j using Lemma 4.2, which required a recalculation of heights and coefficients of the linear form. This is exactly the step responsible for introducing the factor $n!$ and passing from B in (1.3) to B^* in (1.4).

§ 21. Proof of the corollaries

To prove Theorem 2.2, we reduce it to Theorem 2.1 by eliminating the dependent logarithms. We may assume without loss of generality that r is the maximal number of independent logarithms.

If $r = 1$, then some $\alpha \in \mathbb{K}^*$ satisfies

$$\ln \alpha_j = \pm m_j \ln \alpha, \quad m_j \in \mathbb{N}, \quad 1 \leq j \leq n,$$

where $A = H_{\kappa}(\alpha) \leq \min\{A_j: 1 \leq j \leq n\}$ and $|\Lambda| \geq |\ln \alpha|$. Then the estimate follows directly from Liouville's inequality (as in § 6):

$$|\Lambda| \geq \exp(-D \ln 2 - A).$$

Hence we shall assume that $n \geq r > 1$.

Let us consider the case $r = n$, $b_n = 0$. For example, suppose that $b_{n-1} \neq 0$. By induction, we can apply the estimate (2.3) for $n - 1$:

$$\begin{aligned} -\ln |\Lambda| &\leq C(n-1)C_0D^2A_1 \dots A_{n-1} \ln(1.5eB_{n-1}D \ln(eD)), \\ B_{n-1} &= \max\{|b_j|A_j/A_{n-1}: 1 \leq j \leq n-1\} \geq 1 \end{aligned}$$

(since for $j = n - 1$ we have $|b_{n-1}| \geq 1$). Estimating the component $1/A_{n-1}$ in B_{n-1} by (5.3) and passing to B , we get

$$-\ln |\Lambda| \leq C(n-1)C_0D^2A_1 \dots A_{n-1} \ln((1.5D \ln(eD))^2 eBA_n). \quad (21.1)$$

The problem is whether we can introduce an extra factor $x = A_n \geq 0.16$ into the right-hand side of (21.1) and divide the expression under the logarithm sign by x . If so, the right-hand side of (21.1) is a function of the form $f(x) = ax \ln(b/x)$, which increases on the interval $1 \leq x \leq b/e$ (including A_n), and for $x = 1$ we simply get (21.1). Hence, if $A_n \geq 1$, then the estimate (2.5) is obtained from (21.1) by majorizing the logarithm by $2W_0$ and passing to $C(n) > 4e^2C(n-1)$. If $0.16 \leq A_n \leq 1$, then we replace A_n by 1 under the logarithm sign in (21.1) and introduce an extra factor A_n as we pass to $C(n)$. This also yields (2.5), which completes the proof of the case $b_n = 0$.

Suppose now that $1 < r < n$. We may also assume that $b_i \neq 0$, $i > r$. Otherwise we omit the zero terms in the linear form and the factors $A_i \geq 0.16$ in the estimate, which is compensated by improving the estimate as we pass to $C(n')$ with a smaller number n' of terms. Here the case $b_n = 0$ is similar to the situation when $r = n$. Moreover, using (2.4) for dependent logarithms, we get

$$A_1 \dots A_n \xi_0(r) 0.16^{r-n} D \ln(e^{n+4}D) \geq 1 \quad (21.2)$$

and similar estimates for a smaller number of factors.

The preliminary part of the proof is done as in § 6, using (21.2) instead of (5.1), (5.2). We may also assume that conditions (6.1)–(6.4) hold with (6.2) replaced (according to (21.2)) by

$$B \geq 0.99 \frac{1}{n\xi_0(r)} 0.16^{n-r} C(n) C_0 D,$$

or, using a rough estimate since $r > 1$, by the condition

$$B \geq B_0 = 0.16^n C(n) C_0 D / n. \quad (21.3)$$

Thus we suppose that there are integer relations

$$\Lambda_i = x_{i,1} \ln \alpha_1 + \cdots + x_{i,r} \ln \alpha_r - x_{i,i} \ln \alpha_i = 0, \quad x_{i,i} \neq 0, \quad r < i \leq n.$$

But their coefficients may be larger than required. Therefore we take independent integer relations satisfying the conditions of Lemma 4.3:

$$\Lambda'_i = m_{i,1} \ln \alpha_1 + \cdots + m_{i,n} \ln \alpha_n = 0, \quad 1 \leq i \leq n - r.$$

Since the linear forms Λ'_i have maximal rank, Λ_i are expressed in terms of them. It follows that the matrix of coefficients $(m_{i,j})$ has a non-zero minor with indices $j > r$.

We now construct a new linear form

$$\begin{aligned} \Lambda' &= z_0 \Lambda = z_0 \Lambda - z_1 \Lambda'_1 - \cdots - z_{n-r} \Lambda'_{n-r} \\ &= b'_1 \ln \alpha_1 + \cdots + b'_n \ln \alpha_n \neq 0, \quad z_0, \dots, z_{n-r} \in \mathbb{Z}, \quad z_0 \neq 0, \end{aligned}$$

where $b'_{r+1} = \cdots = b'_n = 0$. To do this, it suffices (by Cramér's rule) to put $z_0 = \Delta$, where Δ is the non-zero minor just mentioned, and z_i are obtained by replacing the i -th row of the minor by the row of coefficients b_j , $j > r$. Since $\Lambda' \neq 0$, we may assume without loss of generality that $b'_r \neq 0$. We put

$$\begin{aligned} B' &= |\mathbf{b}'|_A / A_r = \max\{|b_j| A_j / A_r : 1 \leq j \leq r\} \geq 1, \\ W'_0 &= \ln(1.5eB'D \ln(eD)). \end{aligned}$$

Lemma 21.1. *With this notation and provided that (2.4), (6.1) hold, we have the following inequality for $1 < r < n$:*

$$W'_0 \leq \frac{3}{4} n \ln(1.5eBA_n D \ln(eD)). \quad (21.4)$$

Proof. Use (4.1), Lemma 5.3, Stirling's formula and the inequality $\binom{n}{r} \leq n^r / r!$:

$$\begin{aligned} \prod_{i=1}^{n-r} |\mathbf{m}_i|_A &\leq 2.2 \cdot 3^{r/2} \binom{n}{r}^{1/2} \left(\frac{9}{8} D^{3/2} \ln(eD) \right)^r \frac{(r/2)!}{\pi^{r/2}} A_1 \dots A_n \\ &\leq 4\sqrt{n}(nR)^{r/2} A_1 \dots A_n, \quad R = D^3 \ln^2(eD). \end{aligned} \quad (21.5)$$

To estimate the coefficients z_i in the representation of the form Λ' , apply Lemma 4.4 with the max-norm for the last $n - r$ coefficients. Then

$$|z_0| \leq n^{(n-r)/2} \frac{|\mathbf{m}_1|_A \dots |\mathbf{m}_{n-r}|_A}{A_{r+1} \dots A_n}, \quad (21.6)$$

whence

$$|z_0 b_j| A_j \leqslant B A_n n^{(n-r)/2} \frac{|\mathbf{m}_1|_A \cdots |\mathbf{m}_{n-r}|_A}{A_{r+1} \cdots A_n}.$$

We similarly estimate $|z_i m_{i,j}| A_j$, $1 \leqslant i \leqslant n-r$, $1 \leqslant j \leqslant r$. Therefore,

$$B' = |\mathbf{b}'|_A / A_r \leqslant B A_n n^{1+(n-r)/2} \frac{|\mathbf{m}_1|_A \cdots |\mathbf{m}_{n-r}|_A}{A_r A_{r+1} \cdots A_n}.$$

We now apply inequality (21.5). Then one factor A_n is cancelled. We estimate $A_1 \cdots A_{n-1}$ in the numerator by (6.4), A_r in the denominator by (5.3), and the other A_j in the denominator by $A_j \geqslant 0.16$ ($r < j < n$). We get

$$\begin{aligned} B' &\leqslant B A_n n^{1+(n-r)/2} 4\sqrt{n}(nR)^{r/2} \frac{A_1 \cdots A_{n-1}}{A_r \cdots A_{n-1}} \\ &\leqslant 4 B A_n n^{(n+3)/2} R^{(n-1)/2} 0.16^{3-n} \frac{1.5D \ln(eD) B n}{0.99 C(n) D^2 C_0 W_0} \\ &= f_0(B, A_n, D, n). \end{aligned}$$

To complete the proof of (21.4), let us show that

$$f_1(B, A_n, D, n) = (1.5e B A_n D \ln(eD))^{3n/4} \geqslant f_0.$$

We note that the factor A_n occurs with power 1 in f_0 and with power $3n/4 \geqslant 2$ in f_1 . Hence, to verify that $f_1 > f_0$, we must take the minimal possible $A_n = 0.16$. We must also take the minimal possible value for B (which is determined from condition (21.3)) and substitute B_0 in W_0 . Then the dependence of f_1/f_0 on D takes the form

$$\frac{(D^2 C_0 \ln(eD))^{3n/4} D^2 C_0 W_0}{(C_0 D)^2 D \ln(eD) (D^3 \ln^2(eD))^{(n-1)/2}}.$$

Since $C_0 > \ln(eD)$ and $n \geqslant 3$, this expression is an increasing function of D . Hence we may always assume that $D = 1$.

After these substitutions for B , A_n , D , we get

$$\begin{aligned} f_1/f_0 &\geqslant (1.5e 0.16^{n+1} C(n) C_0/n)^{3n/4-2} \\ &\quad \times 0.99 \frac{1}{4} n^{-(n+5)/2} 0.16^{n-2} C(n) C_0 W_0 \\ &\geqslant (0.5 n^{-2} 0.16^n C(n) C_0)^{3n/4-2} \\ &\quad \times 0.24 n^{n/4-9/2} 0.16^{n-2} C(n) C_0 W_0 \geqslant 1, \end{aligned}$$

where each line is $\geqslant 1$ due to the growth of $C(n)$. This proves Lemma 21.1.

In Λ' there appears an extra factor z_0 , which is estimated with the help of (21.5), (21.6) similarly to the estimate of B' :

$$\begin{aligned} |z_0| &\leqslant n^{(n-r)/2} 4\sqrt{n}(nR)^{r/2} \frac{A_1 \cdots A_n}{A_{r+1} \cdots A_n} \\ &\leqslant 4\sqrt{n} n^{n/2} (D^3 \ln^2(eD))^{r/2} 0.16^{r-n} A_1 \cdots A_n \\ &\leqslant e^{2(n-1)C_0} e^{\Omega/e} \leqslant e^{2n\varepsilon_1 G} \end{aligned}$$

(ε_1 , G from § 12). We apply Theorem 2.1 (with n replaced by r) to Λ' . Using (21.4) and the inequality $A_j \geq 0.16$ (for $j > r$), we get

$$\begin{aligned} -\ln |\Lambda| &\leq \ln |z_0| - \ln |\Lambda'| \leq 2n\varepsilon_1 G + C(r)D^2 A_1 \dots A_r C_0 W'_0 \\ &\leq 2n\varepsilon_1 G + \frac{3}{4}nC(n-1)D^2 A_1 \dots A_{n-1} C_0 \ln(1.5eBA_n D \ln(eD)) \\ &\leq \frac{1}{6}nC(n)D^2 A_1 \dots A_n C_0 \ln(1.5eBD \ln(eD)), \end{aligned}$$

which proves (2.5) in the case of dependent logarithms. Here we treat the coefficient A_n as in the case $r = n$, $b_n = 0$, and we apply the inequality

$$2n\varepsilon_1 C(n) + \frac{3}{4 \cdot 0.16}nC(n-1) \leq \frac{1}{6}nC(n).$$

To prove Corollary 2.3, note that we may assume $W_0 \leq 2\ln(eB)$ by (21.3). Hence it suffices to verify the following inequality for $n \geq 2$, $\varkappa \leq 2$:

$$2C(n, \varkappa)C_0(n, D) \min\{1, n/6\} \leq C_1(n, \varkappa) \ln(eD). \quad (21.7)$$

By the obvious inequality $C_0(n, D) \leq C_0(n, 1) \ln(eD)$, it suffices to verify (21.7) for $D = 1$. The main dependence of $C(n)$ on n is of the form $(4e^2)^n$, where $4e^2 = 29.5 \dots < 30$. Hence (21.7) holds for n large. If n is small, we verify (21.7) by direct calculation. This proves (2.6).

To replace B by the B^* in (1.4), it suffices to assume (changing the order of summands if necessary) that A_n is the largest of the A_j . (If all A_j satisfy (2.4), they become equivalent.) Then $B \leq B^*$, which yields the last assertion of the corollary.

§ 22. Appendix

In this section we prove some results of a geometric nature, mainly concerning previous papers of the author.

The estimate of the volume of the truncated set $\mathcal{W}_0$ in § 10 was obtained in [1] by the Brunn–Minkowski theorem. This theorem was also used in the author's papers [5], [8]. Since its application there was indirect and not completely obvious, we want to state explicitly and prove the corresponding assertion.

Proposition 22.1 (The Brunn–Minkowski theorem [21], Theorem 21.1). *Let K_1 , K_2 be compact convex solids in $\mathbb{R}^d$. Then the following inequality holds for all $\lambda_1, \lambda_2 > 0$, $(\lambda_1 + \lambda_2 = 1)$:*

$$\text{Vol}_d^{1/d}(\lambda_1 K_1 + \lambda_2 K_2) \geq \lambda_1 \text{Vol}_d^{1/d} K_1 + \lambda_2 \text{Vol}_d^{1/d} K_2.$$

We used the following result.

Lemma 22.2. *Suppose that $\mathcal{V}$ is a d -dimensional subspace of $\mathbb{R}^n$ ($1 \leq d < n$), $\mathcal{W}$ is a compact convex solid in $\mathbb{R}^n$, $\mathcal{V}^\perp$ is the orthogonal complement to $\mathcal{V}$, and $f(\mathbf{v}) = \text{Vol}_d^{1/d}(\mathcal{W} \cap (\mathbf{v} + \mathcal{V}))$, $\mathbf{v} \in \mathcal{V}^\perp$. Then the function $f(\mathbf{v})$ is convex upwards on the set $\mathcal{W}''$ which is the projection of $\mathcal{W}$ on $\mathcal{V}^\perp$.*

If the convex solid $\mathcal{W}$ is $\mathbf{0}$ -symmetric and if $\mathbf{v} \in \mathcal{V}^\perp$, $\mathbf{v} \neq 0$, then $f(t\mathbf{v})$ is a decreasing function of $t \geq 0$.

Proof. Suppose that $K(\mathbf{v}) = \mathcal{W} \cap (\mathbf{v} + \mathcal{V})$, $\mathbf{v} \in \mathcal{V}^\perp$, and K' is the projection of K on $\mathcal{V}$. Since the projection preserves convexity, linear relations between sets and d -dimensional volumes of solids disposed along $\mathcal{V}$, Theorem 22.1 says that for $\lambda_1, \lambda_2 > 0$ ($\lambda_1 + \lambda_2 = 1$) we have

$$\begin{aligned} \lambda_1 f(\mathbf{v}_1) + \lambda_2 f(\mathbf{v}_2) &= \lambda_1 \text{Vol}_d^{1/d} K(\mathbf{v}_1) + \lambda_2 \text{Vol}_d^{1/d} K(\mathbf{v}_2) \\ &= \lambda_1 \text{Vol}_d^{1/d} K'(\mathbf{v}_1) + \lambda_2 \text{Vol}_d^{1/d} K'(\mathbf{v}_2) \\ &\leq \text{Vol}_d^{1/d} (\lambda_1 K'(\mathbf{v}_1) + \lambda_2 K'(\mathbf{v}_2)) = \text{Vol}_d^{1/d} (\lambda_1 K(\mathbf{v}_1) + \lambda_2 K(\mathbf{v}_2)) \\ &\leq \text{Vol}_d^{1/d} K(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2) = f(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2). \end{aligned}$$

The last inequality holds because $\mathcal{W}$ is convex.

For a symmetric solid $\mathcal{W}$, the function $f(t\mathbf{v})$ is convex upwards in the domain $f(t\mathbf{v}) > 0$ and even in t . This yields the second assertion of the lemma.

In §10 we used Lemma 22.2 to prove that if we cut off the edges of the convex solid $\mathcal{W}$, then the remaining solid $\mathcal{W}_0$ has sufficiently large volume.

Proof of Lemma 5.3. In the general case, (5.4) shows that

$$\|\mathbf{v}\|_2^2 \leq \sum^{(*)} y_\sigma^2 + \left(\sum^{(+)} x_\sigma \right)^2 + \left(\sum^{(-)} x_\sigma \right)^2 \leq (k+2) \|\mathbf{v}\|^2$$

(the sums $\sum^{(\pm)}$ consist of terms of the same sign). Hence, if $\|\mathbf{e}_j\| \leq 1$ for $1 \leq j \leq r$, then $\|\mathbf{e}_j\|_2 \leq (k+2)^{1/2}$ for $1 \leq j \leq r$, and the volume of the parallelepiped they span is at most $(k+2)^{r/2}$. This proves the estimate for the ball $\mathcal{B}(\|\cdot\|, r)$.

To estimate volumes of lattices, it suffices to find a solid $\mathcal{S}_r$ in the space $\mathcal{V}_r = \langle \mathbf{M} \rangle_{\mathbb{R}}$ such that $2\mathcal{S}_r \cap \mathbf{M} = \{\mathbf{0}\}$. Then $\text{Vol}_r \mathbf{M} \geq \text{Vol}_r \mathcal{S}_r$ by Minkowski's theorem.

In the case of $\mathbb{Q}$ we take the solid

$$\mathcal{S} = \left\{ \mathbf{v} \in \mathcal{E}_1(\mathbb{Q}^+): |x_\sigma| < \frac{1}{2} \ln p_\sigma, \quad \sigma > 1 \right\},$$

and if we can take negative numbers in $\mathbb{Q}$, we add the condition

$$|y_1| < \frac{1}{2} \pi.$$

Then the intersection of $\mathcal{S}$ with an r -dimensional subspace is a solid of volume at least $P_1 \dots P_r$ (by Vaaler's lemma on cube sections [22]: see the case of a parallelepiped in [8], Lemma 7 or [23], estimate (3.2)). Moreover, it is clear that $\mathbf{M}_{\mathcal{K}}(\mathbb{Q}) \cap 2\mathcal{S} = \{\mathbf{0}\}$, which yields (5.6).

In the case $D > 1$ we consider the solid

$$\mathcal{S} = \{ \mathbf{v} \in \mathcal{E}_{\mathcal{K}}(\mathbb{K}): \|\mathbf{v}\|_2 < R \}, \quad R = \left(\frac{9}{8} D^{3/2} \ln(eD) \right)^{-1}.$$

We claim that $M_{\mathcal{K}}(\mathbb{K}) \cap 2\mathcal{S} = \{\mathbf{0}\}$. Indeed, let $\mathbf{x} = \mathbf{x}(\alpha)$. If α is not a unit, then some p -adic coordinate satisfies $|x_{\sigma}| \geq \ln 2 > 2R$, whence $\mathbf{x} \notin 2\mathcal{S}$. Hence we assume that α is a unit, that is, $x_{\sigma} = 0$ for $\sigma > D$.

Small algebraic numbers are known for $D = 2, 3$. If α is not a root of unity, then for $D = 2$ we have $\max |x_{\sigma}| > 0.48$ (the minimum is achieved at a root of $x^2 - x - 1 = 0$), and for $D = 3$ we have $\max |x_{\sigma}| > 0.28$ (the minimum is achieved at a root of $x^3 - x - 1 = 0$). In both cases we have $\max |x_{\sigma}| > 2R$. If $\alpha \neq 1$ is a root of unity, then $|\ln \alpha| \geq \pi/3$. In all cases we have $\|\mathbf{x}(\alpha)\|_2 > 2R$ and $M_{\mathcal{K}}(\mathbb{Q}) \cap 2\mathcal{S} = \{\mathbf{0}\}$.

If $D \geq 4$ and $\mathbf{x}(\alpha) \in 2\mathcal{S}$, then $x_1^2 + \dots + x_D^2 < (2R)^2$. Hence $|x_1| + \dots + |x_D| < 2R\sqrt{D}$ and $Dh(\alpha) < R\sqrt{D}$. Moreover, $|\ln \alpha| = |z_1| < 2R \leq R\sqrt{D}$. Hence $H_{\mathcal{K}}(\alpha) < R\sqrt{D}$, and $\alpha = 1$ by Proposition 5.2. Considering $\mathcal{S}_r = \mathcal{S} \cap \langle M \rangle_{\mathbb{R}}$, we now get a ball of radius R whose volume is given by the right-hand side of inequality (5.5).

Remark. Lemma 5.3 yields a working formula for estimating the coefficients in relations between logarithms by Lemma 4.2. We note that the factor corresponding to the volume of the unit ball was overlooked in the corresponding estimates in [8] (Theorems 4 and 6), which were obtained by combining Lemma 4.2 and Minkowski's theorem. Theorem 5 of [8] is true because it is analogous to the case $\mathbb{Q}^+$ of Lemma 5.3, where one must now enumerate prime ideals of the field instead of prime numbers. A corrected version of Theorem 4 of [8] was stated in [23] (without explicit working formulae).

Note that we proved Lemma 5.3 by taking the usual ball, whose sections are also balls, with well-known volume formulae. The proof of the corresponding lemma in [2] actually uses the cube inscribed in the ball, which leads to a weaker estimate.

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